

Solid-Solid Phase Transformations

$$r^* = \frac{-2\gamma T_m}{\Delta H_f \Delta T}$$

r^* = critical radius

γ = surface free energy

T_m = melting temperature

ΔH_f = latent heat of solidification (fusion)

$\Delta T = T_m - T$ = supercooling

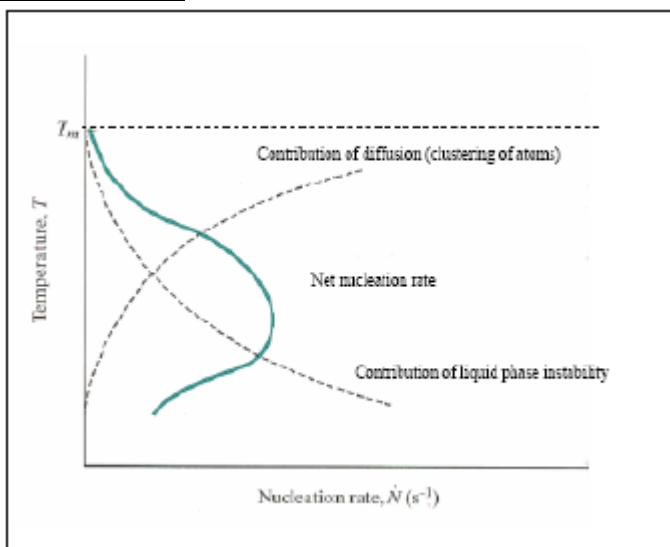
Note: ΔH_f and γ are weakly dependent on ΔT

$\therefore r^*$ decreases as ΔT increases

For typical ΔT $r^* \sim 10$ nm (hundred atoms)

When we reach r^* , the new phase will grow; If the nucleus doesn't reach r^* , it will disgregate and no new phase will grow from that point.

Nucleation rate

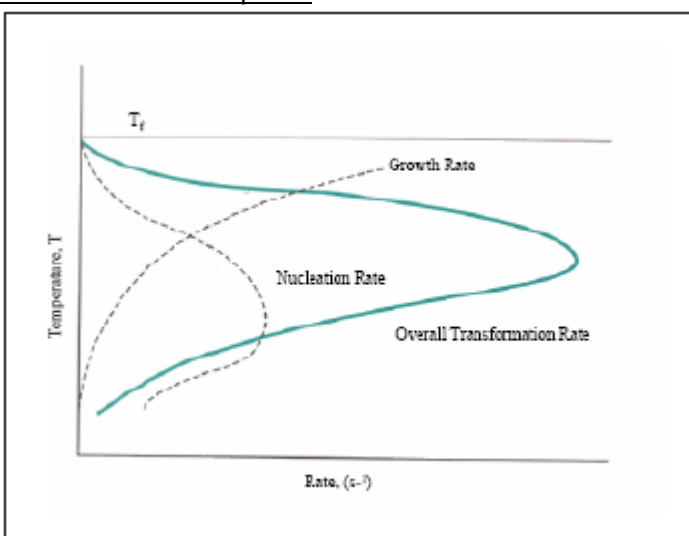


The overall nucleation rate is a product of two opposite working factors:

- The driving force for solidification **increases** with T decrease;
- While the diffusion that is responsible for the clustering process **decrease** with T decrease.

These explain the "knee-shaped" nature of the transformation curve with maximum at some value below T_f .

Growth rate of new phase



The overall transformation rate is a product of two factors:

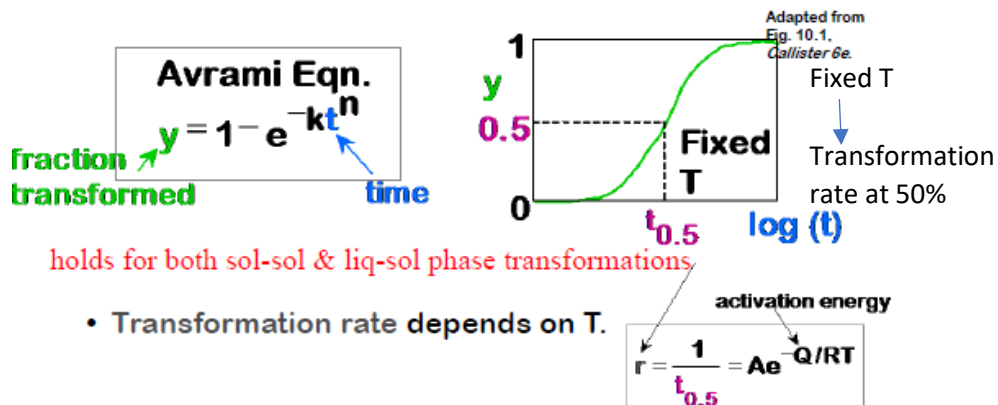
- **Nucleation**;
- **Growth**, which is thermally activated process, typically described by Arrhenius function.

Diffusion contributes to the expansion of the area of the nucleus.

PT rate achieves **maximum** at some T below T_f !

Isothermal Transformation Growth Kinetic

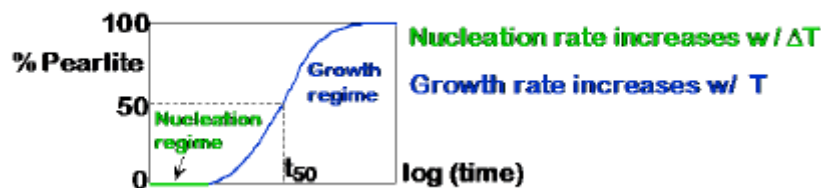
- **volume (fraction) transformed** depends on time.



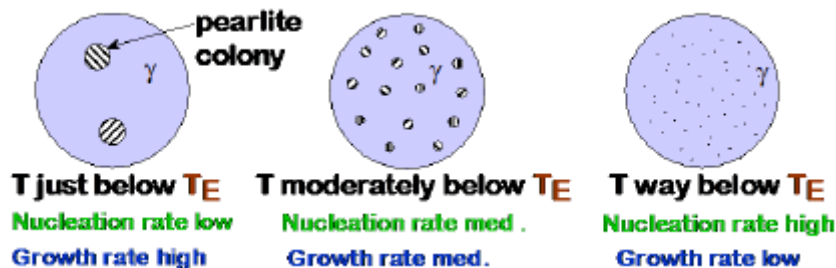
Very high and severe quenching $\rightarrow$ high supercooling and high cooling rate $\rightarrow$ high number of nuclei $\rightarrow$ with growth the particles start to expand and whenever the nuclei touch each other they will stop growing $\rightarrow$ very fine microstructure.

Isothermal Phase Change (ex. Pearlite formation)

- Reaction rate is a result of **nucleation** and **growth** of crystals.

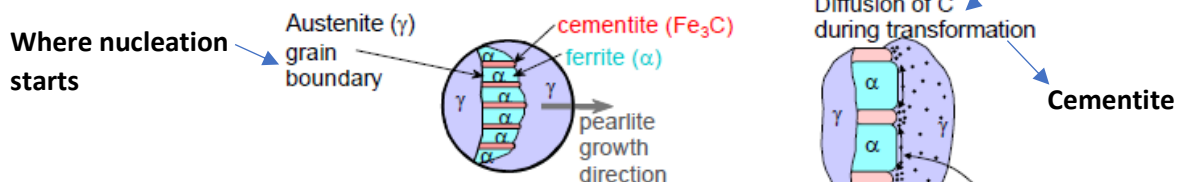


- Examples:



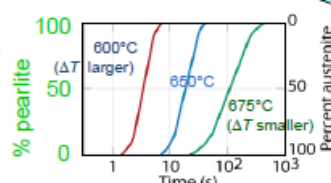
Ex. Pearlite Growth on slow cooling

- Transformation of austenite to pearlite:



- For this transformation, rate increases with (ΔT) $[T_{\text{eutectoid}} - T]$.

$A_1, 727^\circ C$



Pearlite can be grown at different temperatures

Coarse pearlite $\rightarrow$ formed at **higher temperatures** – relatively soft

Fine pearlite $\rightarrow$ formed at **lower temperatures** – relatively harder

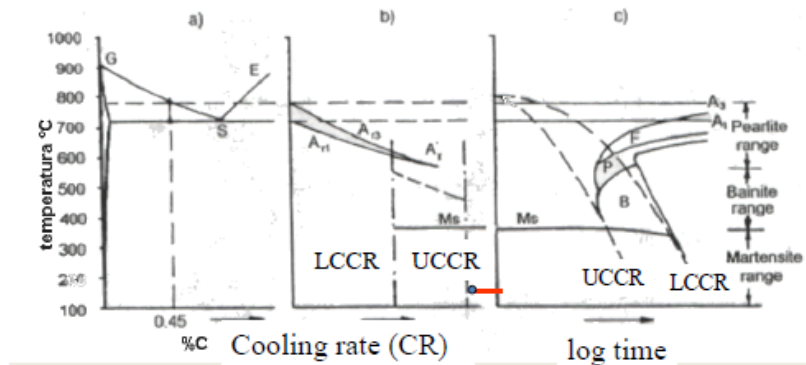
High temperature $\rightarrow$ lower undercooling ΔT (@675°C on graph)

Low temperature $\rightarrow$ Larger undercooling ΔT

The longer the nucleation time, the finer the microstructure and the higher the number of nucleation sites.

A small number of nucleation sites $\rightarrow$ higher growth rate, and vice versa.

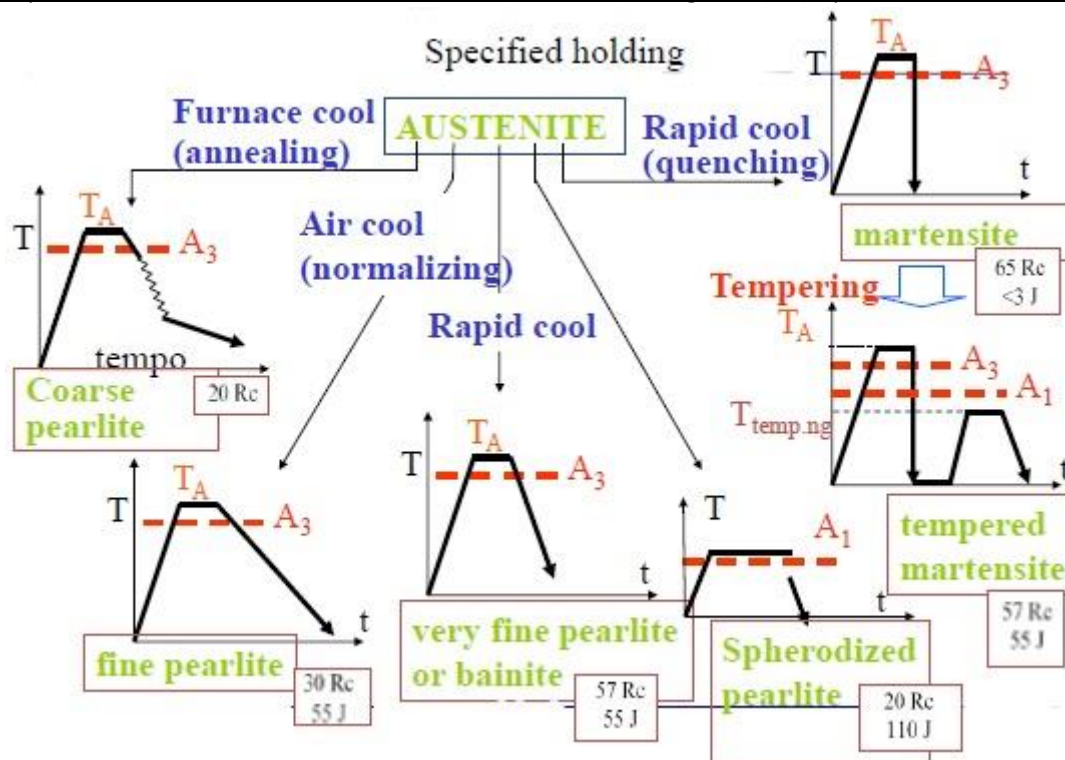
Non-equilibrium phase change on cooling: influence of cooling rate: ex. AISI 1045 steel



a) Equilibrium metastable Fe-Fe₃C phase diagram; b) transformation temperatures change with increasing CR; c) CCT diagram for AISI1045 steel.

What happens during the phase transformation when the cooling rate increases $\rightarrow$ transformation takes longer time and at low temperature as the cooling rate increases. As the cooling rate increases, temperature decreases and at a certain point A1 is exactly equal to A3 $\rightarrow$ we are losing the pearlitic formation, some nuclei will grow instead of pearlite. Critical cooling rate in the lower range $\rightarrow$ some new phases will start to grow in place of pearlite (Bainite or Martensite, depending on T).

Relationship between Microstructure, Hardness, Resilience, cooling rate (CR) upon Heat Treatment



Resilience is the absorption of impact energy.

Coarse pearlite is steel in the softest condition (low hardness).

Spheroidal pearlite has ferrite as matrix with spheroids of carbide and pearlite. It is highly ductile.

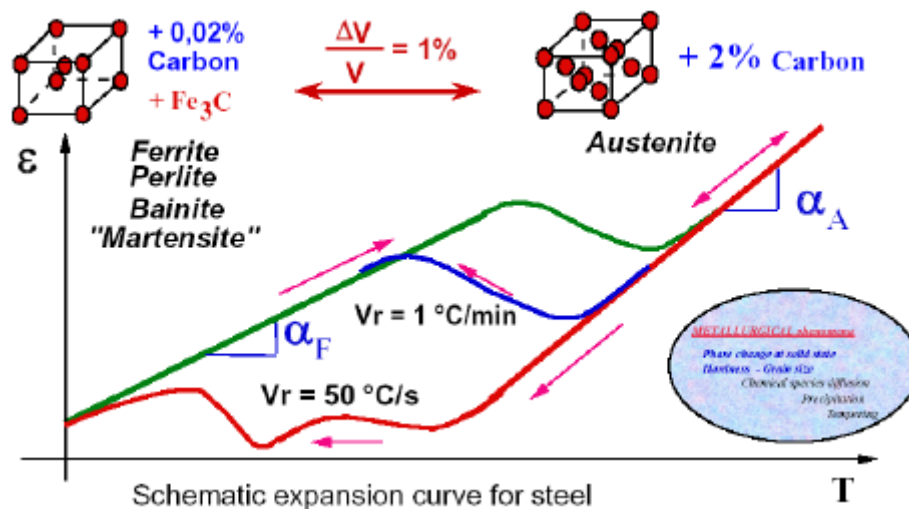
Tempering requires quenching as the first step.

Sharp cooling results in **martensite**.

Low hardness $\rightarrow$ higher ductility; Larger hardness $\rightarrow$ lower toughness.

Diffusion may occur at a high temperature in a shorter time or at low temperature in a longer time.

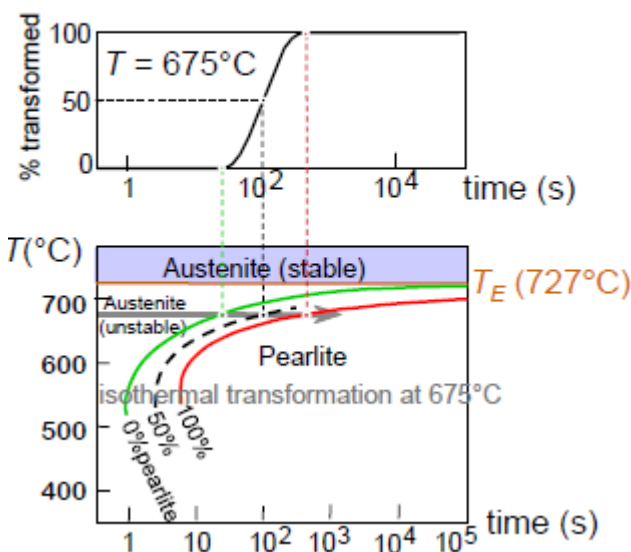
Volume change during phase transformations



Ferrite will transform into austenite by changing the number of atoms $\rightarrow$ change of volume. Increase of temperature leads to the expansion of the material, strain decreases and the phase changes.

Non-equilibrium Isothermal Transformation

- The Fe-Fe₃C system, for $C_0 = 0.76 \text{ wt\% C}$
- A transformation temperature of 675°C .



2 solid curves are plotted:

- one represents the time required at each temperature for the start of the transformation;
- the other is for transformation completion.
- The dashed curve corresponds to 50% completion.

The austenite to pearlite transformation will occur only if the alloy is supercooled to below the eutectoid temperature (727°C).

Time for process to complete depends on the temperature.

In continuous transformations, the curves are shifted to right and lower part $\rightarrow$ higher time and lower temperature.

TTT diagram $\rightarrow$ we start from the austenitic state and we cool down and see what happens in the graph.

Instantaneous quenching $\rightarrow$ material will start to nucleate; phase transformation $\rightarrow$ change of volume $\rightarrow$ the new phase does not have enough time to adapt to the change of volume $\rightarrow$ twinning helps it to adapt $\rightarrow$ phase transformation controlled by shear.

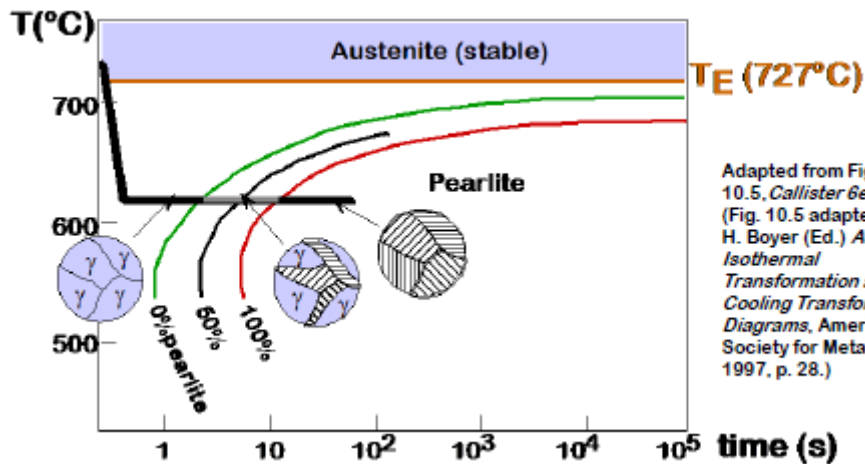
Martensite $\rightarrow$ no nucleation, only growth; formed by the shear (twinning), not by the diffusion.

Bainite $\rightarrow$ most difficult to be obtained by the continuous cooling. Obtained with a mixture of salts: at the melting point of salts, it will not change its state (constant temperature).

[TTT (Time-Temperature-Transformation) diagram $\rightarrow$ plots of temperature vs time, generated from the percentage transformation vs time measurement and are useful for understanding the transformation of alloy steel at elevated temperatures]

Ex: Cooling history Fe-C alloy

- Eutectoid composition, $C_o = 0.77\text{wt}\%C$
- Begin at $T > 727^\circ\text{C}$
- Rapidly cool to 625°C and hold isothermally.



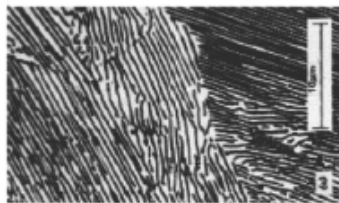
Pearlite microstructure

It occurs at a different cooling rates.

Two cases:

Small undercooling

- T_{transf} just below $T_E(A_1)$
 - Larger T : diffusion is faster
 - Pearlite is coarser.



Adapted from Fig. 10.6 (a) and (b), Callister 6e. (Fig. 10.6 from R.M. Ralls et al., *An Introduction to Materials Science and Engineering*, p. 361, John Wiley and Sons, Inc., New York, 1976.)



- Smaller ΔT : colonies are larger

- T_{transf} well below T_E
 - Smaller T : diffusion is slower
 - Pearlite is finer.

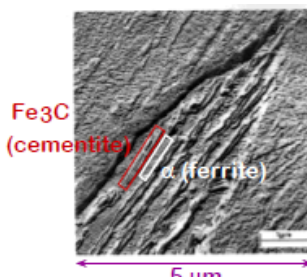
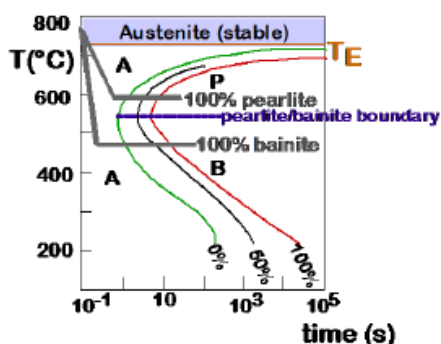


- Larger ΔT : colonies are smaller

Thinner and closer lamellae

Bainite microstructure

- Bainite:**
 - α lathes (strips) with long rods of Fe_3C
 - diffusion controlled.
- Isothermal Transf. Diagram**



(Adapted from Fig. 10.8, Callister, 6e. (Fig. 10.8 from *Metals Handbook*, 8th ed., Vol. 8, *Metallography, Structures, and Phase Diagrams*, American Society for Metals, Materials Park, OH, 1973.)

Bainite reaction rate:

$$r_{\text{bainite}} = e^{-Q/RT}$$

No colonies (different from pearlite)

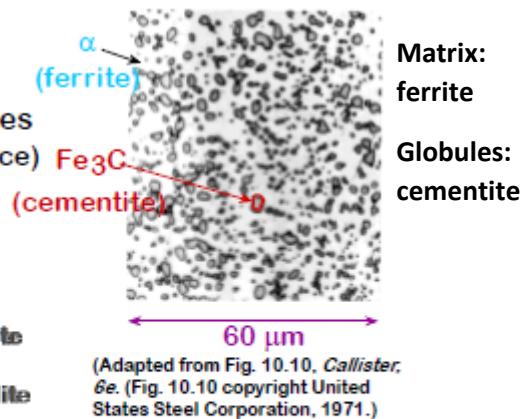
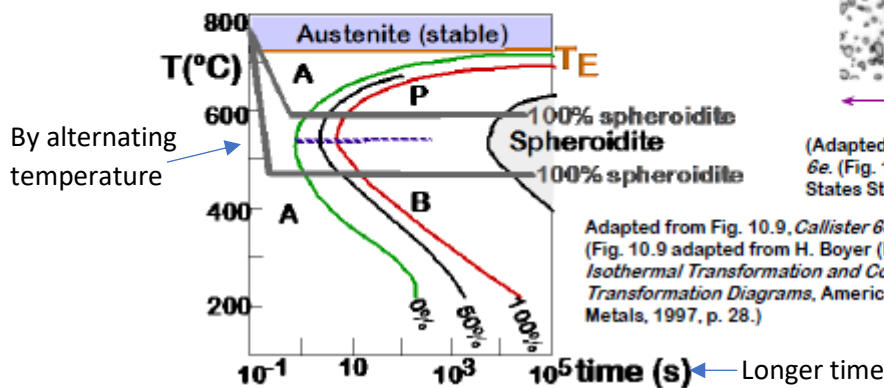
- Smaller and more detailed structure
- ➔ highly resolved microstructure.
- Long nucleation and transformation time.

Spheroidal Carbide

- **Spheroidite:**

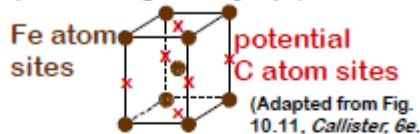
- α crystals with spherical Fe_3C
- diffusion dependent.
- heat bainite or pearlite for long times
- reduces interfacial area (driving force)

- **Isothermal Transf. Diagram**

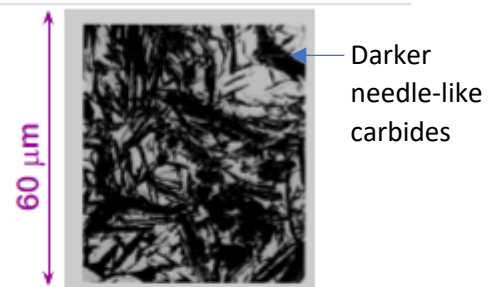
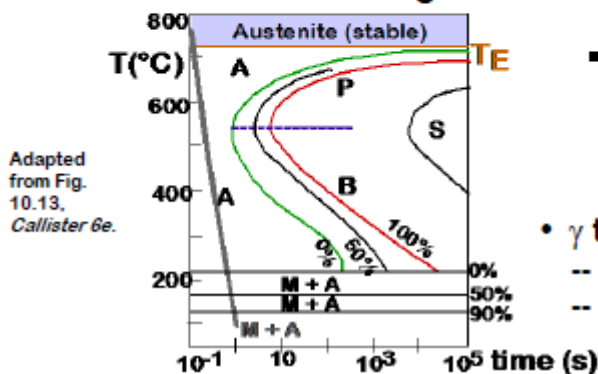


Martensite microstructure

- γ (FCC) to Martensite (BCT)
(involves single atom jumps)



- **Isothermal Transf. Diagram**



Martensite needles
Austenite

(Adapted from Fig. 10.12, Callister, 6e. (Fig. 10.12 courtesy United States Steel Corporation.)

- γ to M transformation..

- is rapid!
- % transf. depends on T only.

Stress applied to Martensite $\rightarrow$ crack propagation because martensite is too brittle.

Martensite is formed in carbon steels by the rapid cooling (quenching) of the austenite form of iron at such a high rate that carbon atoms do not have time to diffuse out of the crystal structure in large enough quantities to form cementite.

The martensitic transformation differs from the other transformations in that it is not time depended and occurs almost instantaneously, the portion of austenite transformed to martensite depends only on the temperature to which it is cooled.

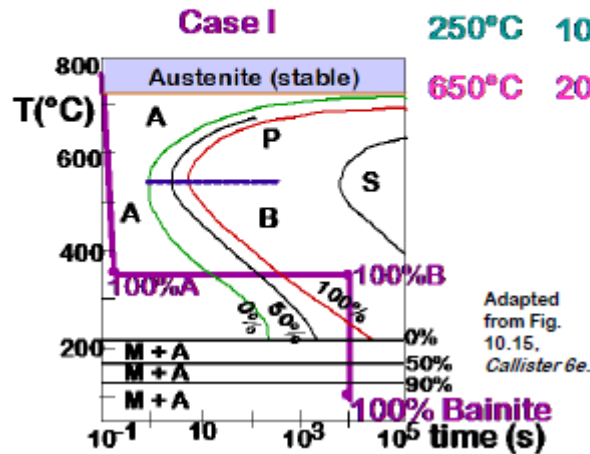
- $C_0 = C_{\text{eutectoid}}$
- Three histories...

Rapid cool to: Hold for: Rapid cool to: Hold for: Rapid cool to:

350°C 10^4 s T_{room} 10^4 s T_{room}

250°C 10^2 s T_{room} 10^2 s T_{room}

650°C 20s 400°C 10^3 s T_{room}



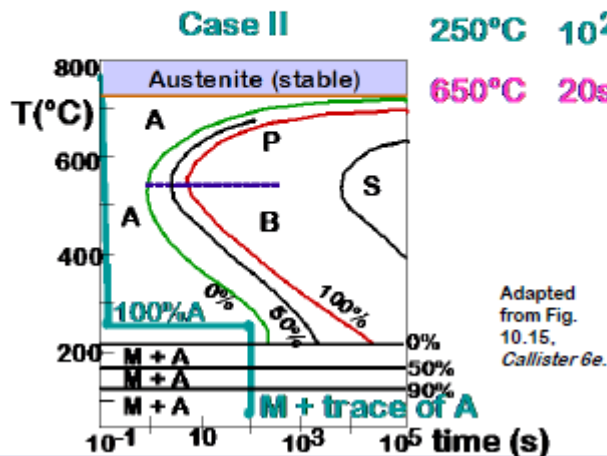
- $C_0 = C_{\text{eutectoid}}$
- Three histories...

Rapid cool to: Hold for: Rapid cool to: Hold for: Rapid cool to:

350°C 10^4 s T_{room} 10^4 s T_{room}

250°C 10^2 s T_{room} 10^2 s T_{room}

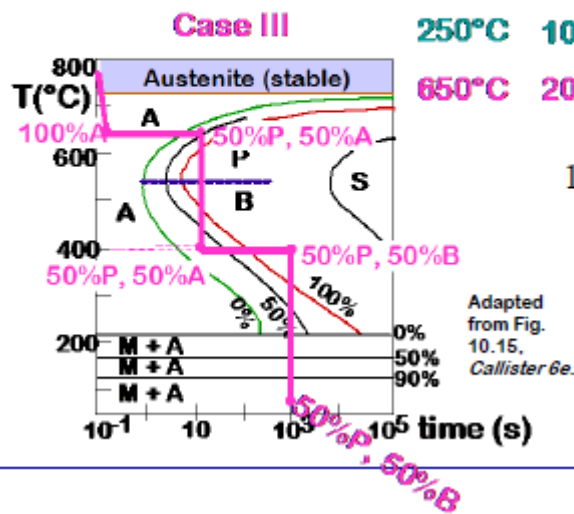
650°C 20s 400°C 10^3 s T_{room}



- $C_0 = C_{\text{eutectoid}}$
- Three histories...

Rapid cool to: 350°C Hold for: 10^4 s T_{room} Rapid cool to: 10^4 s T_{room}
 Hold for: 250°C Hold for: 10^2 s T_{room} Hold for: 10^2 s T_{room}
 Rapid cool to: 650°C Hold for: 20s Rapid cool to: 400°C Hold for: 10^3 s T_{room}

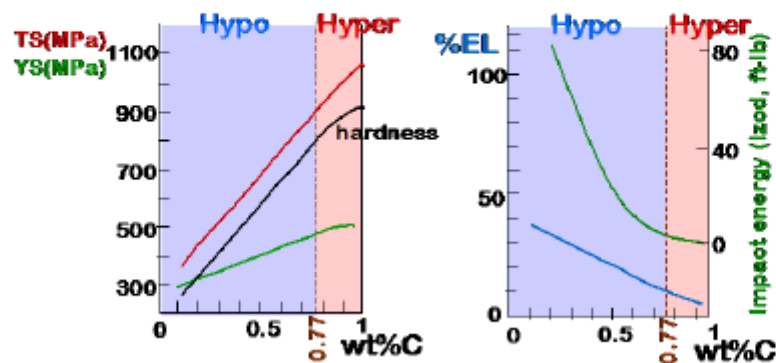
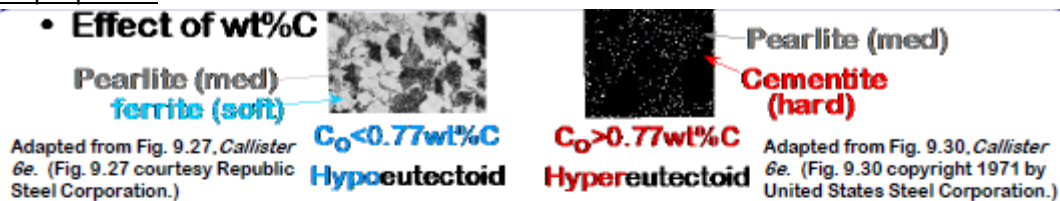
12.5%B + 37.5%A



14

Mechanical properties

- Effect of wt%C

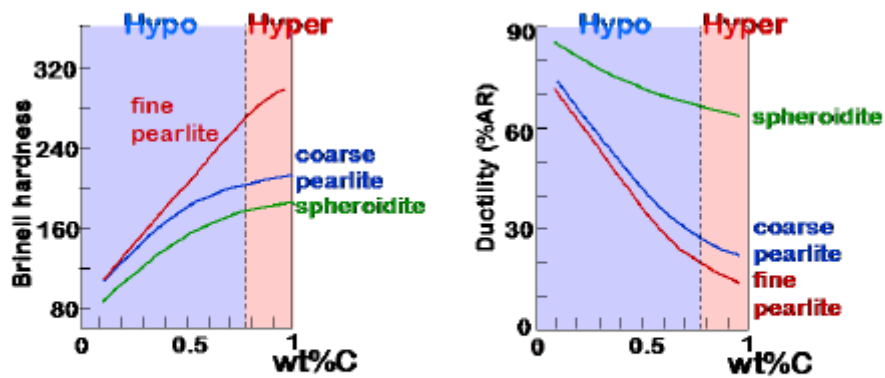


- More wt%C: TS and YS increase, %EL decreases.

Hardness for both hypo and hyper

Elongation as well as the impact energy decreases with increasing %C

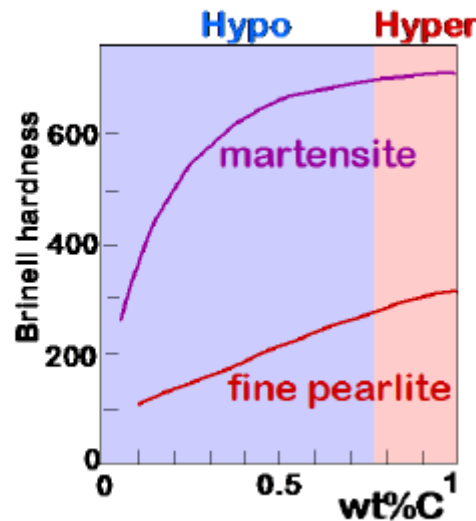
- Fine vs coarse pearlite vs spheroidite



- Hardness: fine > coarse > spheroidite
- %AR: fine < coarse < spheroidite

Adapted from Fig. 10.21, *Callister 6e*. (Fig. 10.21 based on data from *Metals Handbook: Heat Treating*, Vol. 4, 9th ed., V. Masseria (Managing Ed.), American Society for Metals, 1981, pp. 9 and 17.)

- Fine Pearlite vs Martensite:

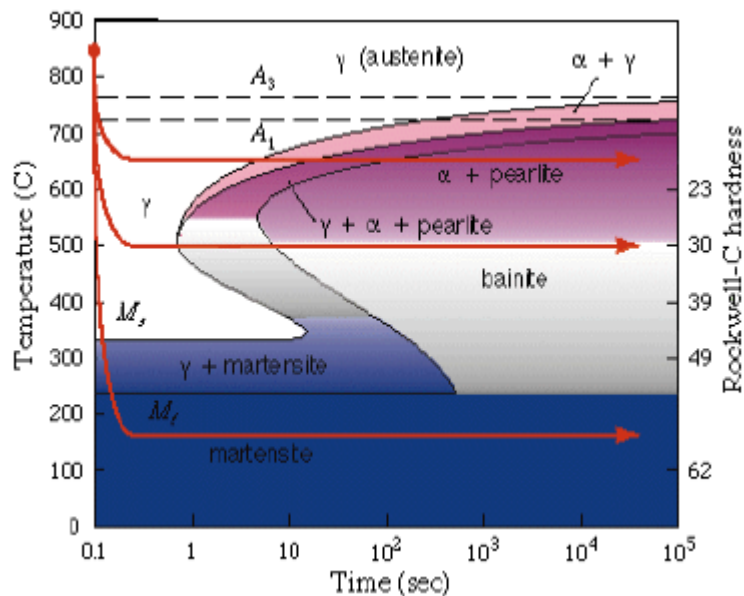


Adapted from Fig. 10.23, *Callister 6e*. (Fig. 10.23 adapted from Edgar C. Bain, *Functions of the Alloying Elements in Steel*, American Society for Metals, 1939, p. 36; and R.A. Grange, C.R. Hribal, and L.F. Porter, *Metall. Trans. A*, Vol. 8A, p. 1776.)

Hardness depends only on Carbon content

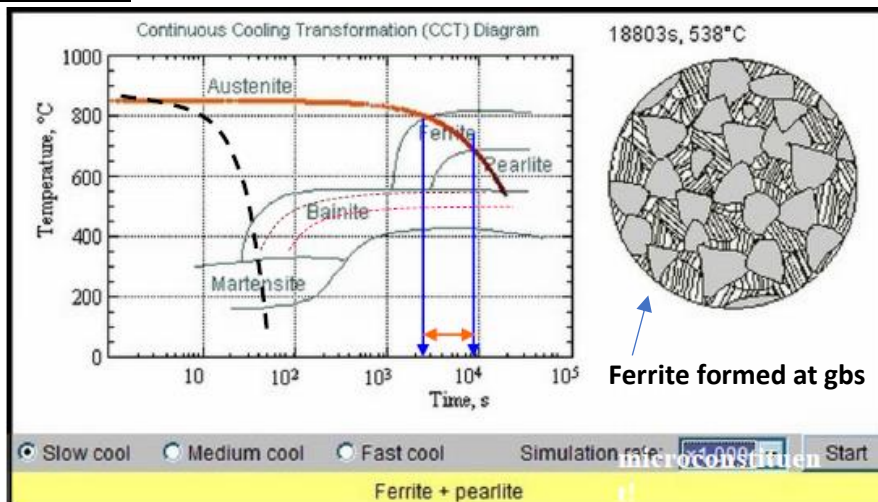
- Hardness: fine pearlite << martensite.

Special (Isothermal) H.T.



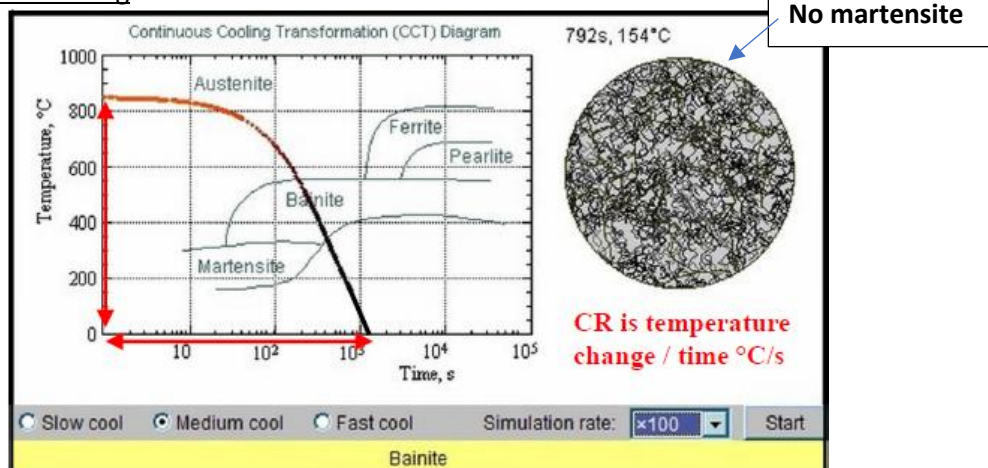
When we have ferrite or cementite, we have to solve the problem because in non-equilibrium diagram (like TTT, CCT) we have only the initial formation of ferrite, the transformation is never completed (it is completed in the phase diagram because we have much time to solve the problem of diffusion). In order to quantify the amount of ferrite approximately, we have to see how much is ΔT or how much is the cooling rate affecting ferrite formation. The larger the undercooling, the lower %ferrite. We apply the lever rule between two T and calculate ΔT .

Slow cooling on CCT curves

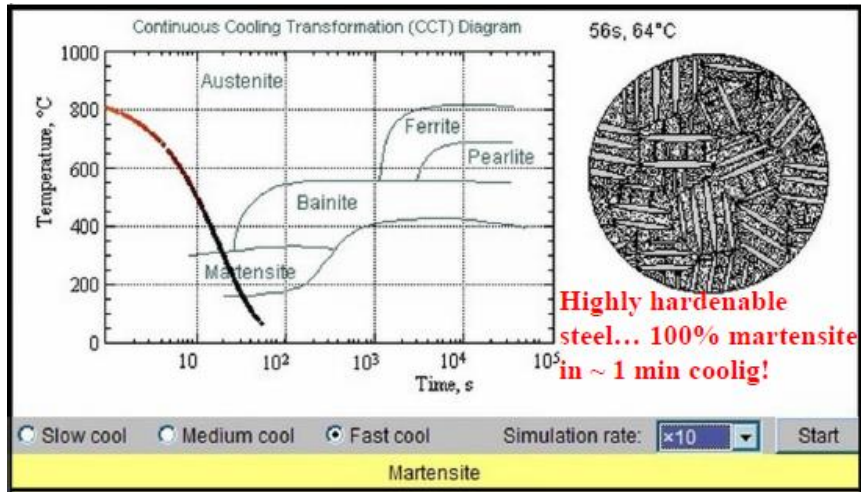


When the cooling rate is very slow (long time of transformation), the cooling curve will intersect Ferrite. While the curve crosses the region of Ferrite, its amount will increase and expand. Then Pearlite will start forming at boundaries between Ferrite and Austenite. There is no end to the formation of Pearlite. Bainite will start growing. The same analysis is applied to the following slides.

CCT curves: moderate cooling



CCT curves Quenching cooling



Higher cooling rate.

Martensite amount increases as the curve penetrates more in its region.

Pearlite microstructure change with the CR

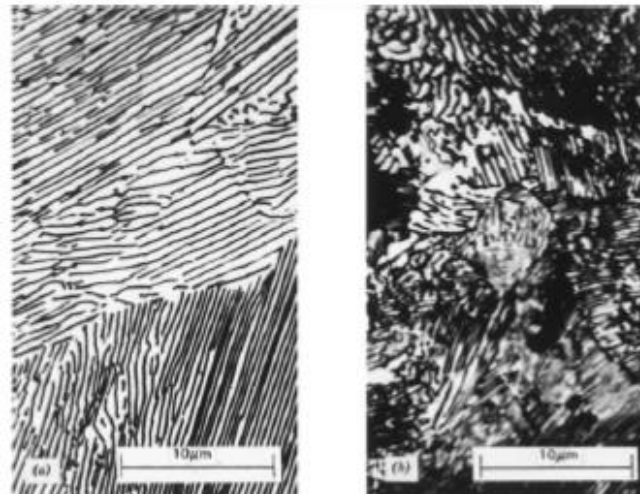
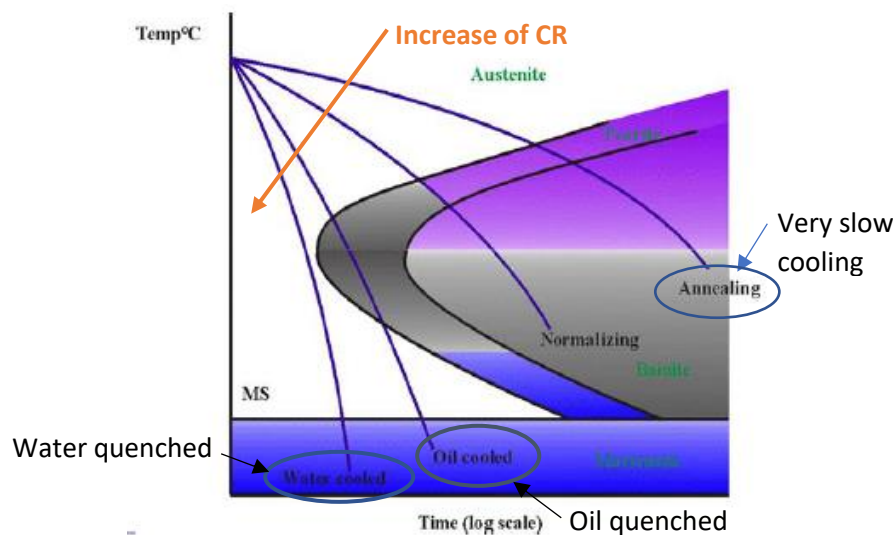
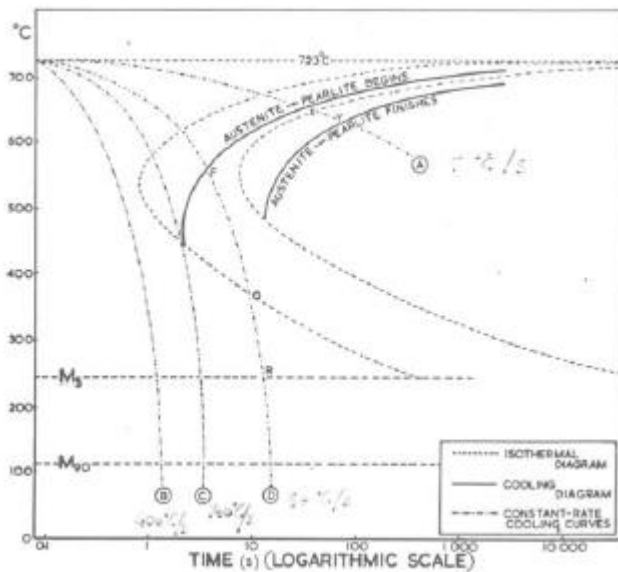


FIGURE 10.6 Photomicrographs of (a) coarse pearlite and (b) fine pearlite. 3000 $\times$. (From K. M. Ralls et al., *An Introduction to Materials Science and Engineering*, p. 361. Copyright © 1976 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

Heat treatment of steels by CCT curves



CCT curves vs TTT curves



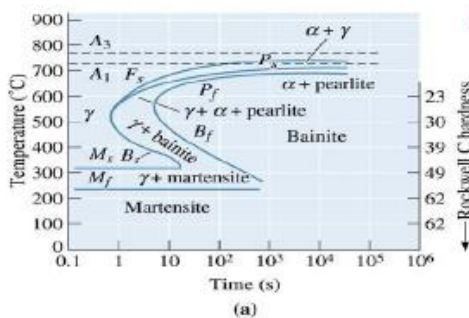
CCT curves are translated bottom-right compared to TTT curves.

Incubation times in continuous transformations are lower compared to isothermal transformations 705 °C (conventional values) → temperature at which most transformations at lower temperature will be controlled.

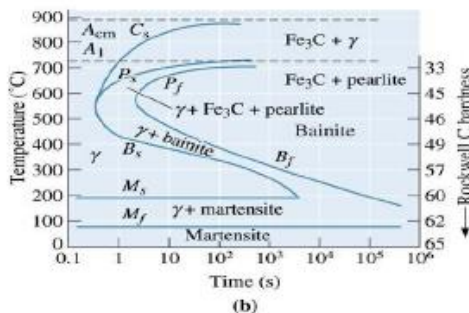
At temperature lower than 705, the curve will follow a diffusion mechanism.

The transformation will be completed when the cooling curve will intersect the beginning and the end of the TTT curves.

TTT curves for hypo- and hyper-eutectoid Steels

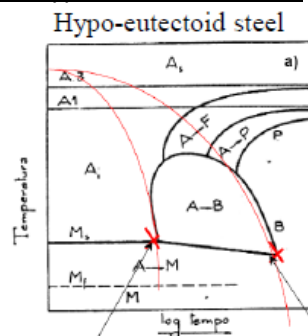


Hardenable steels
For constructions
C=0.5%

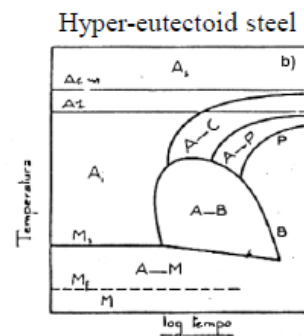


Tool steel
C=1.1%

CCT curves for hypo- and hyper-eutectoid steels



UCCR=
min critical CR to obtain
ONLY martensite at Tamb

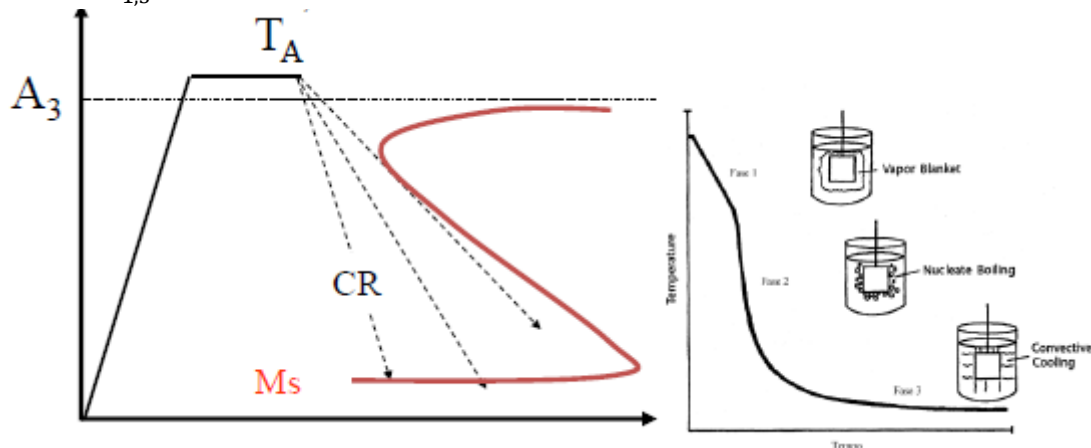


LCCR=
Max critical CR
to avoid martensite at Tamb

Quench hardening

Steel hypoeut.: $A_3 + 50^\circ\text{C}$

Steel eutectoid: $A_{1,3} + 50^\circ\text{C}$

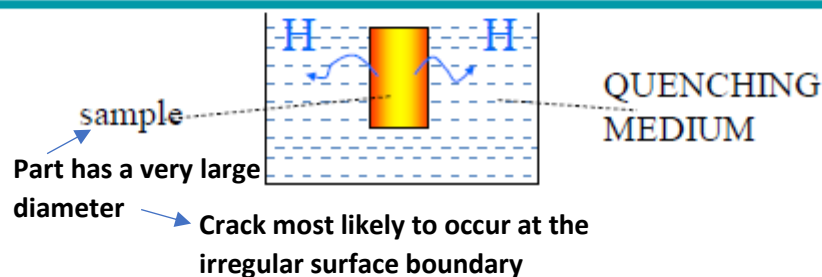


The actual CR at point of workpiece depends on the geometry and mass of the component. Its surface will cool faster than its core.

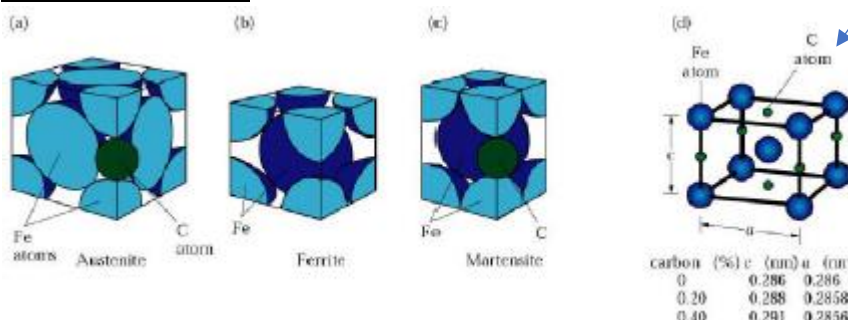
Quench severity of various cooling media

QUENCH SEVERITY

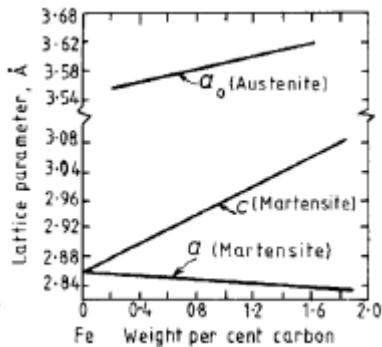
Medium	H Coefficient	Cooling Rate at the Center of a 1-in. Bar ($^\circ\text{C/s}$)
Oil (no agitation)	0.25	18
Oil (agitation)	1.0	45
H ₂ O (no agitation)	1.0	45
H ₂ O (agitation)	4.0	190
Brine (no agitation)	2.0	90
Brine (agitation)	5.0	230



Martensite structure



Note the interstitial position of C atoms and the consequent increase of c lattice constant with increasing %C; these effects cause a lattice distortion of the U.C. of martensite which, from BCC ferrite becomes tetragonal structure.

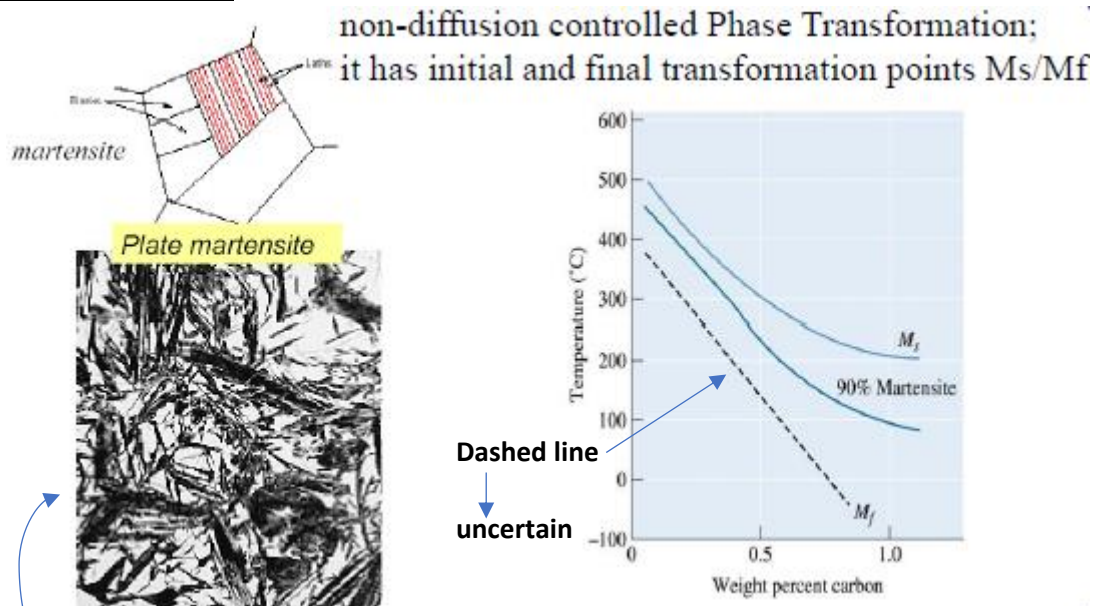


On the ordinate, we have the elongation of the cube.

We have different behaviors → with increasing of %C into the steel, **(a)** will shrink while **(c)** will expand. The cube will expand in one direction by shrinking the transversal lattice constant.

(c) elongation is much larger than the shrinking of **(a)** → it will expand more along z and then along xy.

Martensitic Transformation

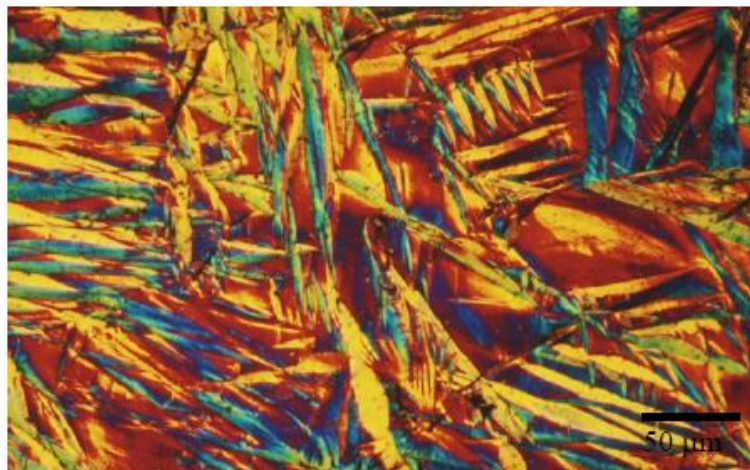


- White region → tetragonal crystal structure (lattice distorted along the c direction)
- Black region → needle-like cementite

Sharp edge morphology of carbides (needle-like cementite) is the reason for brittleness of martensite → stress applied intensifies at the tip of the needles.

Increasing %C will delay the transformation to lower T.

Martensite Transformation

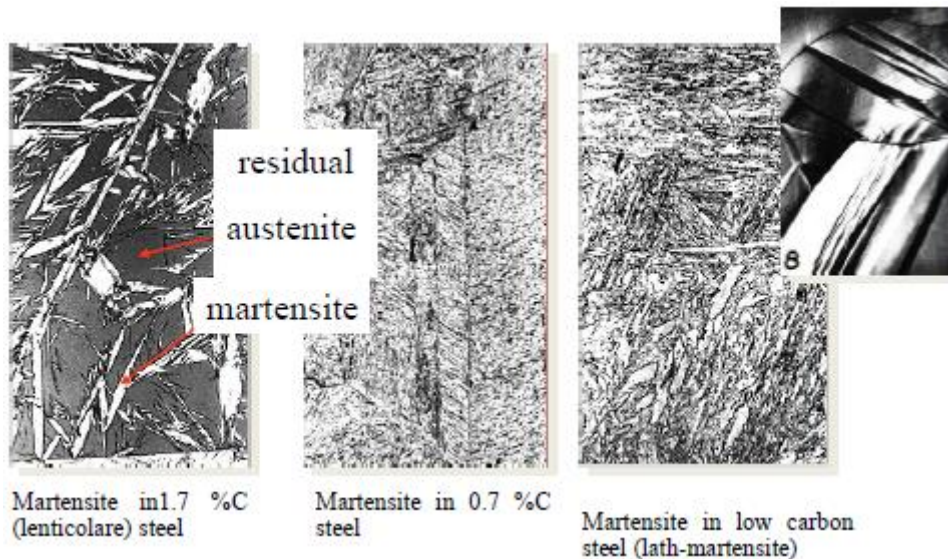


Blue area
↓
Cementite particles

Morphology of Martensite vs %C

It is a highly metastable microstructure!!

Martensite can be formed regardless of the amount of carbon we have, but its microstructure morphology and hardness changes with carbon content.



Residual austenite → some austenite remains after transformation because we cool down very rapidly, but martensite finish is much lower than the temperature of quenching because of the high carbon content.

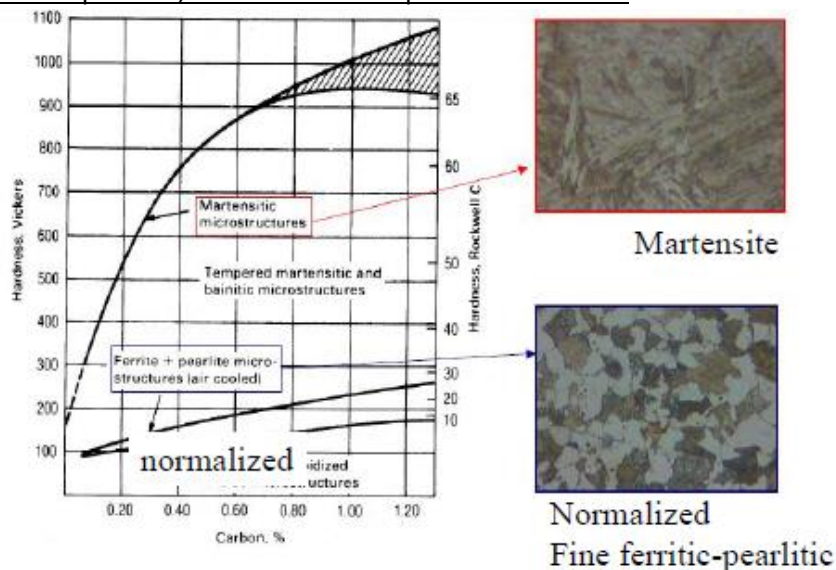
Martensite in 1.7%C (lenticolare) steel: hypo-eutectoid steels → needles are much sharper and very clear, tetragonal structure is much richer of carbon.

Martensite in 0.7%C steel: much finer needles observed at high resolution.

Hardness after Quenching (ex. Eutectoid Steel)

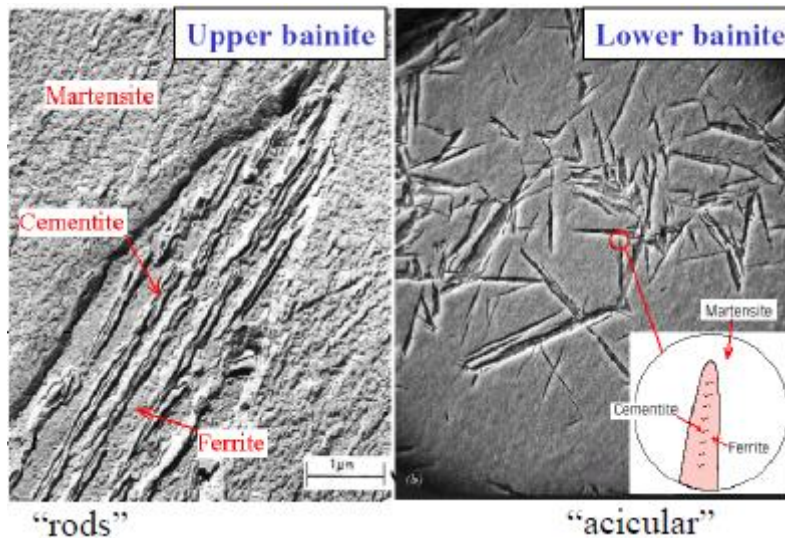
Cooling rate	Microstructure	HRC
Water quenching	Martensite	65 ← Highest hardness
Oil quenching	Pearl. Bain. Mart.	40-45
Furnace cooling	Pearlite	15-20
Austempered	Bainite	40-45

Correlation between composition, microstructure in plain carbon steels



When we quench steels, the hardness of martensite will depend on the carbon content.

Upper and Lower Bainite (500°C – Ms)



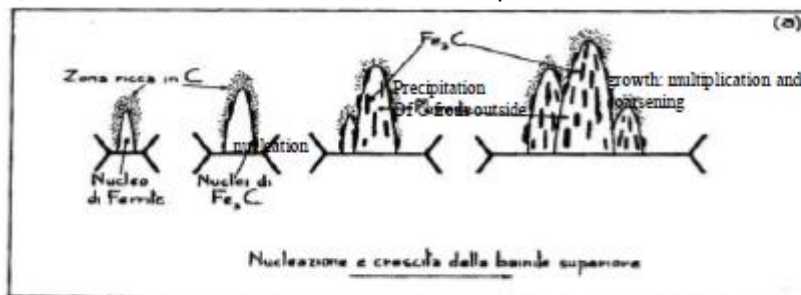
Upper Bainite: the one we obtain at high T; during cooling, it's the first to be formed. It looks like pearlite, but the colonies are squeezed → refined pearlite → higher hardness and higher strengthening.

Lower Bainite: plates as alternating lamellae of finer cementite and ferrite (matrix) → higher hardness and strength. It can be found in Damascus swords.

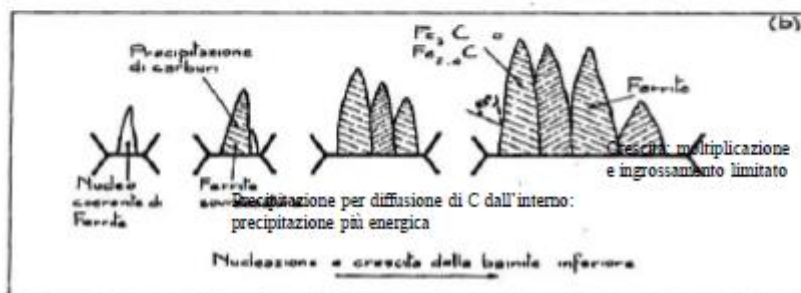
Optimum tradeoff between mechanical strength and toughness.

Excellent alternative to tempered martensite for hardness (tempered steels).

In general, with decreasing temperature during the phase transformation, the microstructure will look like martensite, but it will retain some similarities with pearlite.



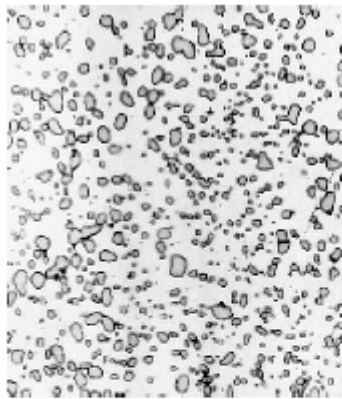
Isolated rods in ferritic matrix
Equilibrium structures are still present



Isolated rods mixed with needle-like carbides ($\alpha+\theta$)
Equilibrium structure are still present ma in lower amount

BAINITE FORMATION: Acicular: ferrite + carbide needles.

Spheroidizing Annealing

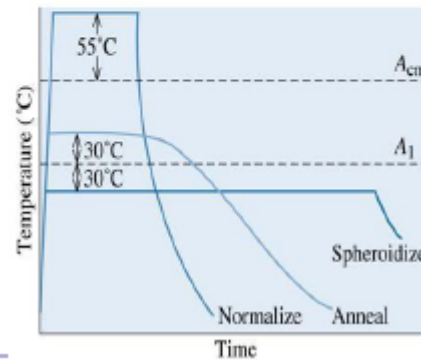


Spheroidized steel; the small particles are Iron carbides; the grey matrix is ferrite 1000x.
Copyright 1971 - United States Steel Co.

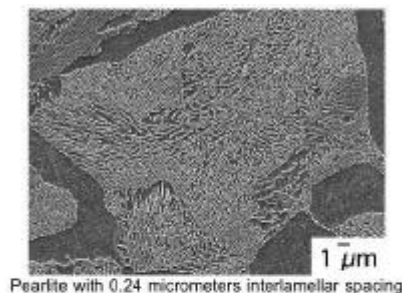
This microstructure in hyper-eutectoid steels promotes the **largest ductility** which **useful in shaping operations**

Prolonged annealing to T just below A_1

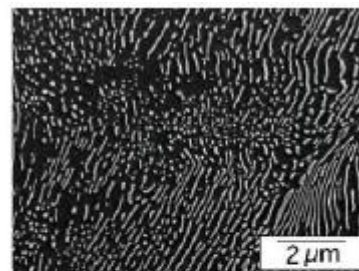
Pearlite or Bainite $\xrightarrow[\text{18 - 24 h}]{\text{heating}}$ spheroidal cementite
 $\approx$ just below A_1
Decrease in interfacial area α/θ



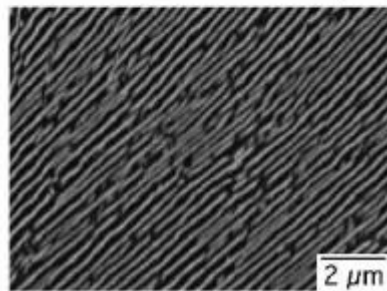
Subsequent steps during spheroidizing Heat Treatment



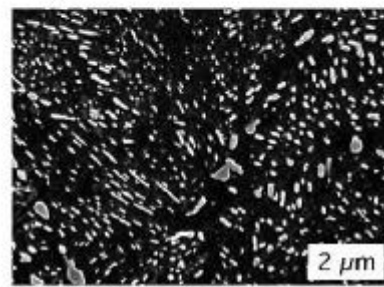
Pearlite with 0.24 micrometers interlamellar spacing



After spheroidisation for 1.5 h at 750 centigrade



Pearlite with 0.13 micrometers interlamellar spacing

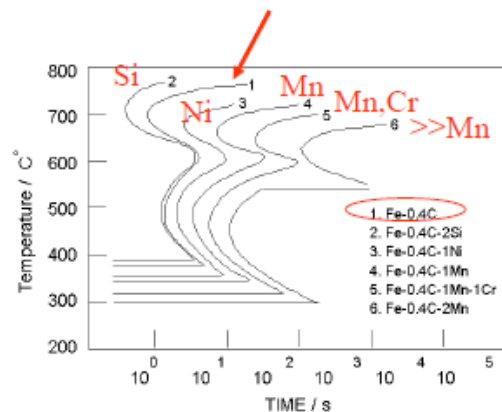


After spheroidisation for 1.5 h at 750 centigrade

Spheroidizing of pearlite occurs by breaking the lamellae into large pieces which will be made spheroidal.

Influence of Alloy Elements on TTT curve

Shape and position of TTT curves is affected



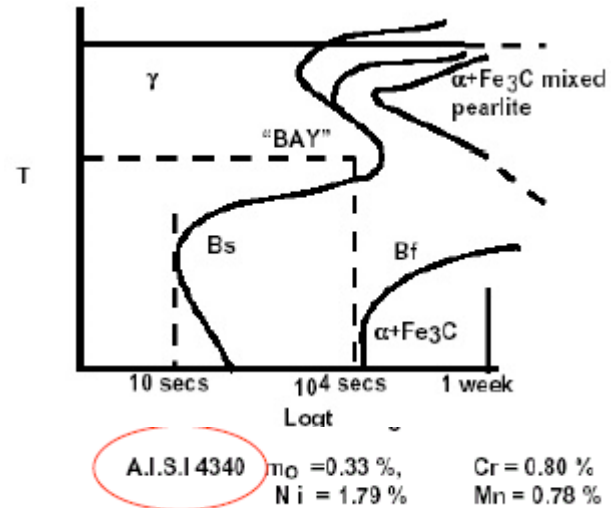
Sometimes we can have a bay between Pearlite and Bainite (created by Mn).
Displacement of the curve to the right $\rightarrow$ longer time.

TTT curve as a function of %alloy elem.: AISI 4340

When %Cr, %Mo, %Ni are small (total content <10%) the a bainitic nose and the so-called bainitic bay appears because B transforms faster at lower T than TE. The larger driving force is due to the larger undercooling which balances the inertia (delay) effect.

$$M_s \text{ } ^\circ\text{C} = 539 - 423 (\%C) - 30.4 (\%Mn) - 17.7 (\%Ni) - 12.1 (\%Cr) - 7.5 (\%Mo)$$

Example of empirical relation

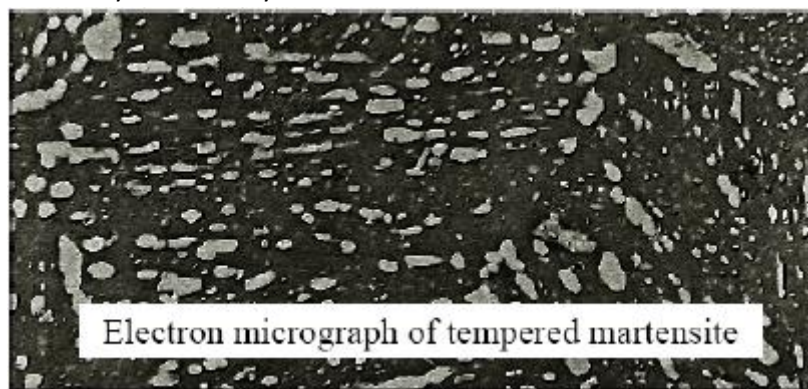


AISI 4340: used for very critical applications, it has the largest compromise between strength and toughness

What is Tempered Martensite?

It is the final step after obtaining martensite.

- Tempered martensite is obtained when Fe_3C is let precipitate from the undercooled Martensite; it is composed of ferrite and cementite.
- Morphology is not markedly acicular any more



The needles have disappeared and became more rounded (reason of increasing toughness).

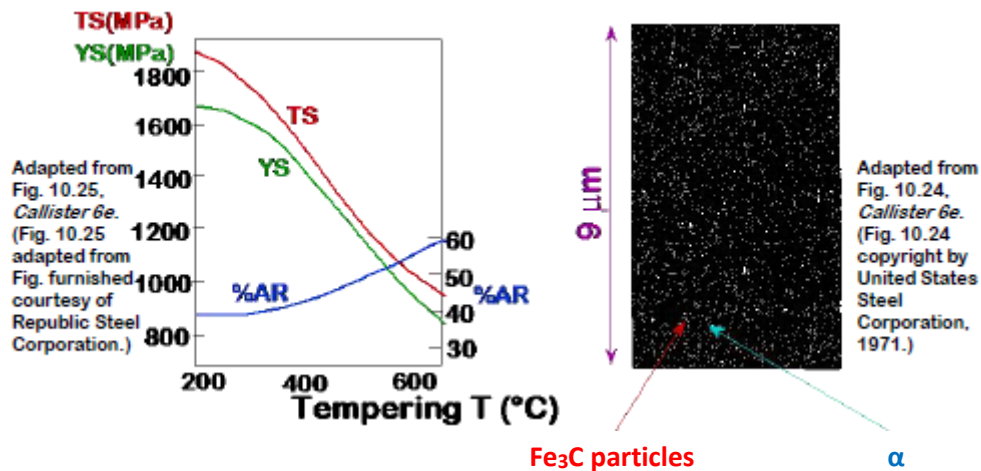
Martensite is so brittle that it needs to be modified for practical applications. This is done by heating it to 250-650°C for some time (tempering) which produces **tempered martensite**, an extremely fine-grained and well dispersed cementite grains in a ferrite matrix.

- Tempered martensite is less hard/strong as compared to regular martensite but has enhanced ductility (ferrite phase is ductile).
- Mechanical properties depend upon cementite particle size: fewer, larger particle means less boundary area and softer, more ductile material – eventual limit is spheroidite.
- Particle size increases with higher tempering temperature and/or longer time (more C diffusion) – therefore softer, more ductile material.

Tempering improves the hardness and the elasticity of steel by reheating and then cooling it.

Tempering martensite

- Reduces brittleness of martensite;
- Reduces internal stress caused by quenching;
- Produces extremely small Fe_3C particles surrounded by α ;
- Decreases TS, YS but increase %AR.



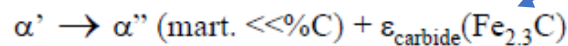
We have different strength and elongation value with varying T.
A higher temperature will result in lower strength (TS, σ_s) and higher residual elongation (%).

Tempering steps

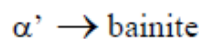
Process in which martensite microstructure will change depending on temperature.

Tempering = quench hardening + heating below A1

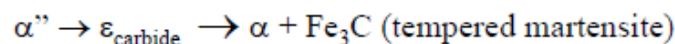
1^o step 80-160 °C



2^o step 230-280 °C



3^o step 160-400 °C

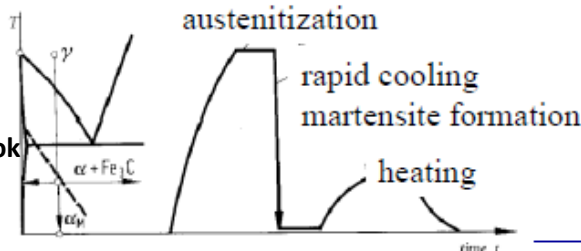


4^o step 400-700 °C

→ Evolution towards equilibrium ($\alpha + p$)

At this T, we are almost losing the effect of quenching

Microstructure will look like combination of Ferrite and Pearlite



Tetragonal martensite changing into α' which will change into α'' (α' after losing carbon thanks to diffusion)

Carbon will go to precipitate as $\epsilon_{\text{carbide}}$

Metastable before having Fe₃C, it's essential

Increase of temperature → we are promoting diffusion → carbon is escaping from the crystal lattice.

Tempering of carbon steels

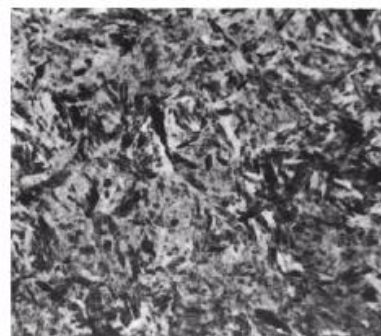
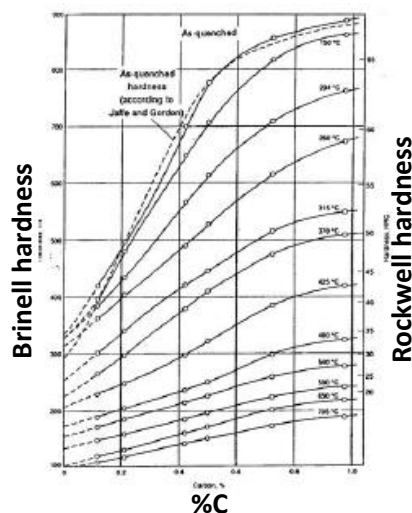
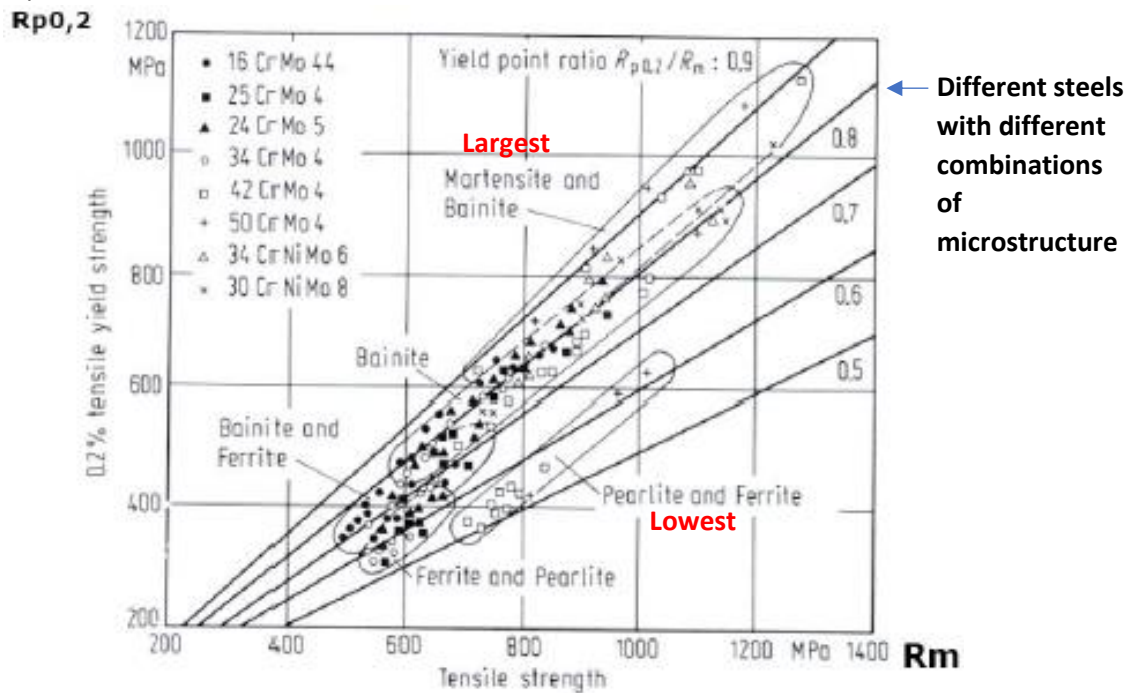


Figure 10.18 The microstructure of tempered martensite, although an equilibrium mixture of α -Fe and Fe₃C, differs from those for pearlite (Figure 9-2) and bainite (Figure 10-9), 825 $\times$. This particular microstructure is for a 0.50 wt % C steel comparable with that described for Figure 10-16. (From Metals Handbook, 8th Ed., Vol. 7: Atlas of Microstructures, American Society for Metals, Metals Park, Ohio, 1972.)

The graph shows the mechanical properties when we apply tempering to different carbon steels. When the temperature is high, hardness is low; high hardness is obtained right after quenching.

Mechanical properties vs Microstructure



Alloying Effects on Hardenable Steels

- **C**: contributes to $\uparrow$ (Ys, HRC) but $\downarrow$ Kv (toughness) and DBTT (Ductile to Brittle Transition Temperature), $\uparrow$ hardenability (H); forms carbides but not very hard and not so fine compared to (Cr, Ti, Ta, W, V, Mo)
- **Mn**: contributes to $\uparrow$ (Ys, HRC); its role depends on %C; with $\uparrow$ Mn, $\downarrow$ ductility and weldability. Mn has strong effect on hardenability ($\uparrow\uparrow$ H); strong austenitizer after Ni
- **P**: $\uparrow$ (Rs, HRC) but $\downarrow$ Kv (toughness).. with $\uparrow$ %C, $\downarrow\downarrow$ (Kv, $\Delta L\%$) in tempered steels.
- **S**: in the form of sulfides will $\downarrow$ (Kv, $\Delta L\%$) especially in the transversal direction; with $\uparrow$ %S, $\downarrow$ weldability
- **Al**: powerful deoxydizer of melt, grain refiner, deleterious for content $>0.1\%$
- **Si**: main deoxydizing element of steels. Less effective than Mn in $\uparrow$ HRC, UTS of plates and coils. $\uparrow$ Si, detrimental effect on surface finishing in low alloy steels; $\uparrow$ elastic limit and $\uparrow$ magnetic properties.
- **Cu**: in large content is detrimental in hot worked steels since it $\downarrow\downarrow$ surface finish. If $\text{Cu} > 0.20\%$, $\uparrow$ Rcorr
- **Ni** ferrite strengthening: $\uparrow$ UTS keeping high (Kv, $\Delta L\%$); does not form carbides (stays into solid solution); with $\uparrow$ Ni, $\uparrow\uparrow$ Kv or $\downarrow\downarrow$ DBTT; strong austenitizer
- **Mo**: $\uparrow$ Rcreep in low alloy steels at high T; $\uparrow\uparrow$ H. $\downarrow$ DBTT, solves the temper -embrittlement problem of low alloy steels; $\uparrow$ resistance to high Temperature
- **B**: $\uparrow\uparrow\uparrow$ H
- **Cr, Ti, Ta, W, Nb, V, Si, Mo**: grain refiners because form very fine and finely distributed carbides at GBs, $\uparrow\uparrow$ HRC without depressing toughness; $\uparrow$ resistance to high T; $\uparrow$ resistance to wear.

Role of Alloy Elements in Tempered Steels

Hardenability	UTS	Toughness	Workability.
Mn	Co	Mn	S
Mo	Mn	Ni	P
Cr	Mo	Mo	Mn
C	Cu	Nb	Se
B	Cr	Ta	Te
Ti	C	Zr	Pb
P	Ni	Ce	
	Nb	Cr	
	Si	Ca	
	Ta, W, V	Te, V	
	Co		

Weakest carbide formers
Most dissolved in α

↓

Carbide formers

↓

↑

dissolve in α

↑

Mn

Cr

Mo

W

V

Ti

Nb

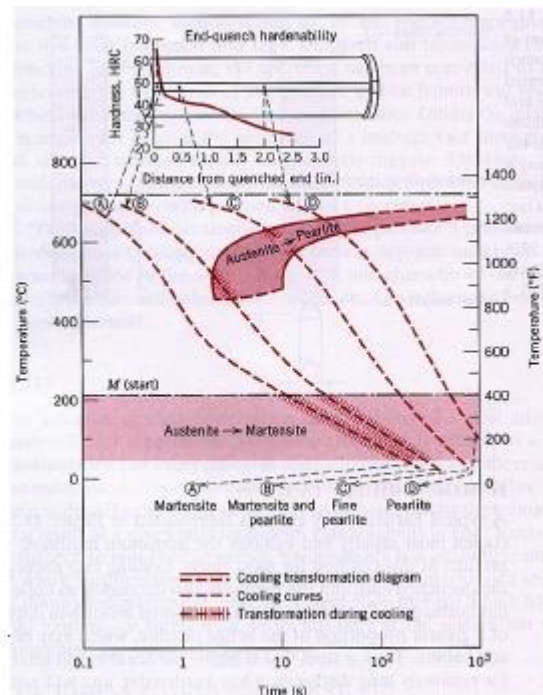
(Cb)

strongest carbide formers
Least dissolved in α

Hardenability and TTT curves



The cooling rate varies with position.



Hardenability is the depth up to which a material is hardened after putting to a heat treatment process. It is a material property, it doesn't depend on the outside system. It is measured through the Jominy test.

Jominy test: a round metal bar of standard size is transformed to 100% austenite through heat treatment and is then quenched on one end with room temperature water. The cooling rate will be highest at the end being quenched and will decrease as the distance from the end increases. Subsequent to cooling, the hardenability is then found by measuring the hardness along the bar. The further away from the quenched end that the hardness extends, the higher the hardenability → decreasing of hardness with increasing of Jominy distance.

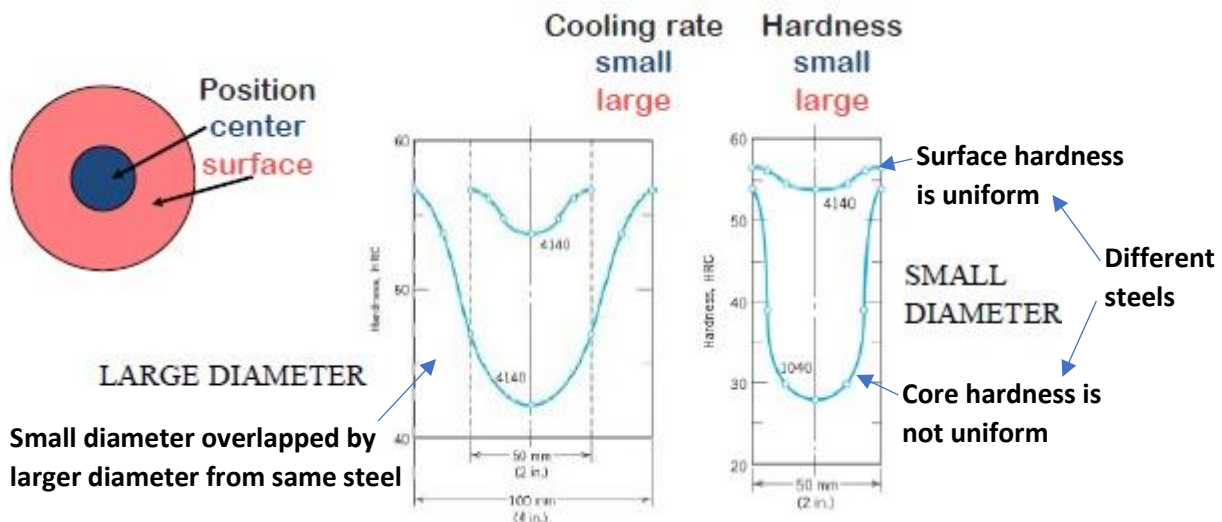
Geometry/diameter effect on quenching

We have steel with C and we want to obtain martensite at the end of quenching; we can't exceed in cooling rate, especially if the component has irregular geometry because it will break.

Normalization doesn't make the steel strong enough → we have to select the tempering temperature and time for the steel to achieve the strength required.

When surface-to-volume ratio increases:

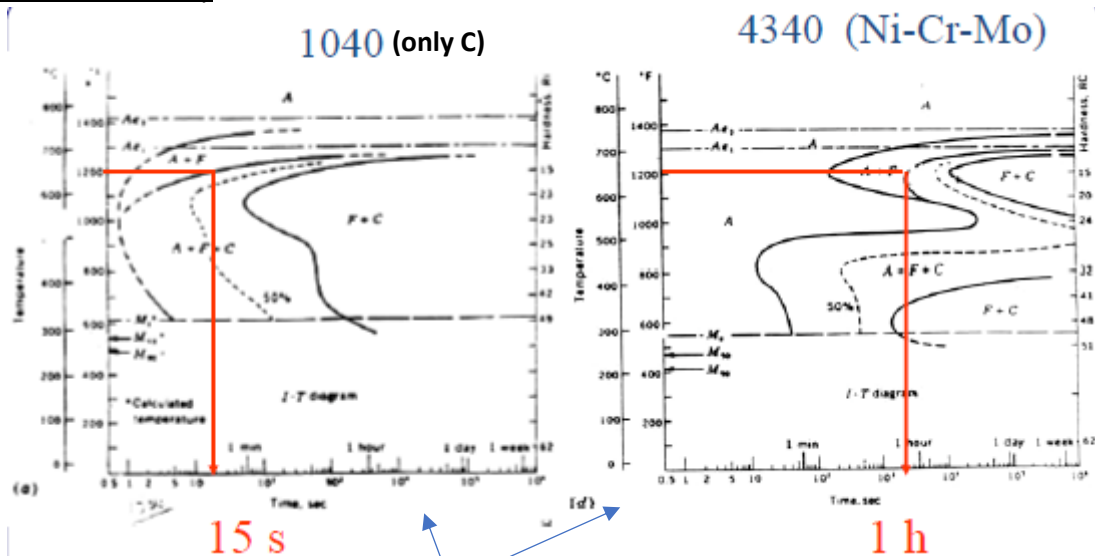
- Cooling rate increases
- Hardness increases



Hardenability allows to obtain martensite up to at least 50% into depth. The problem is to get martensite into the core because the condition of cooling in the core is less effective and the larger the diameter of the material the more difficult it is to obtain the right cooling rate to achieve martensite.

Adding alloy elements makes hardness uniform.

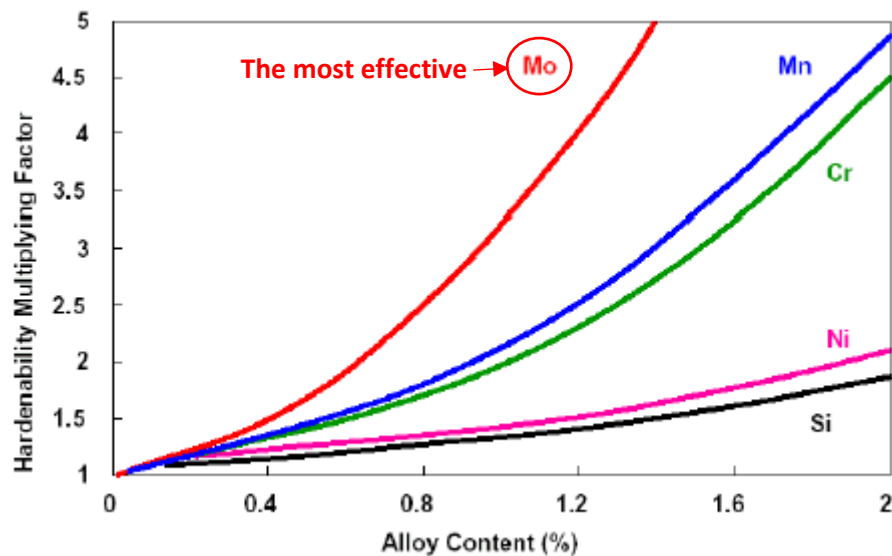
TTT curves: hardenability



Different steels with different TTT diagrams

Same transformation (initial formation of pearlite, 50%) but different time. By adding Ni, we obtain the same product but with different time. Adding alloy elements results in hard products with low cooling rates.

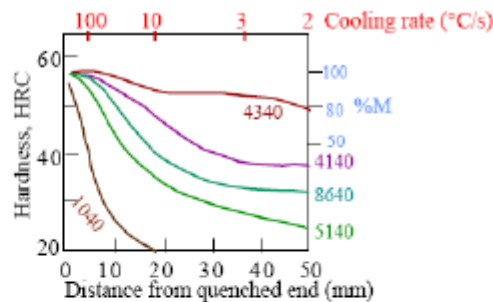
Hardenability vs %alloy elements



Hardenability of Tempered Steels

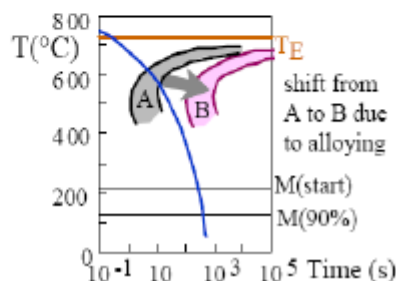
- Jominy end quench results, C = 0.4wt%C

4140: 1.0 Cr / 0.2Mo
4340: 1.85Ni / 0.8Cr/0.25Mo
5140: 0.85Cr
8640: 0.55 Ni/ 0.5Cr/ 0.20Mo



By adding alloy elements, we are shifting the steel curve to the right. We can cool the steel with slower cooling curve and guarantee martensite at any point in the steel.

- "Alloy Steels"
(4140, 4340, 5140, 8640)
--contain Ni, Cr, Mo
(0.2 to 2wt%)
--these elements shift
the "nose".
--martensite is easier to form.



Hardenability: comments on previous fig.

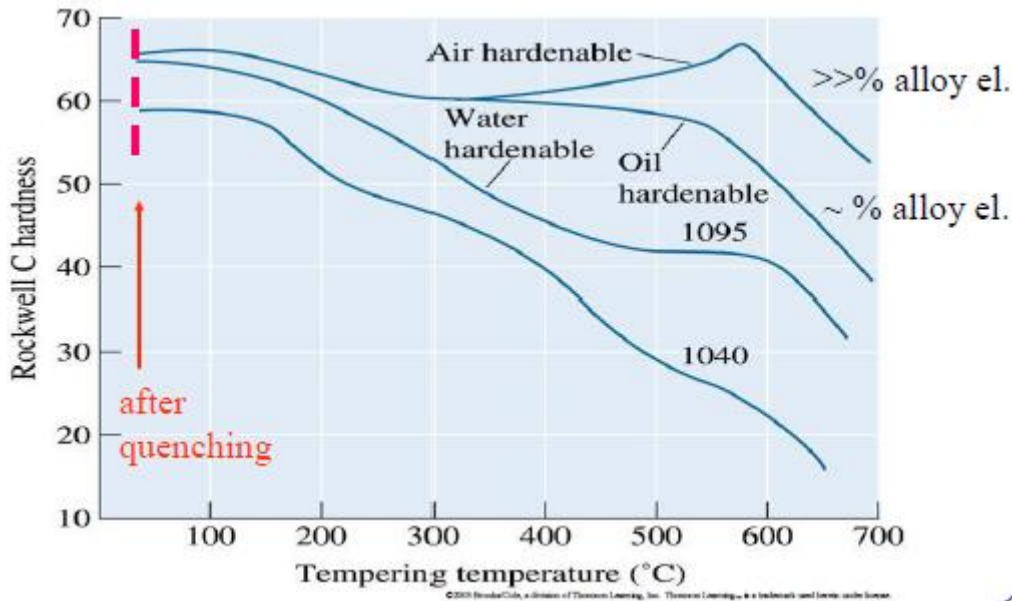
Figure shows 1 reference plain carbon steel (1040), 4 low alloy (hardenable) steels

- All steels have the same hardness at the quenched end (57 HRC); **this is because hardness in steels only depends on %C (which has to completely dissolved into solid solution before quenching).**
- The shape and position (along t) will determine hardenability of the steel.
- **Hardenability increases with the following order:**

1040	<	5140	<	8640	<	4140	<	4340
only C		only Cr		0.55 %Ni 0.50%Cr 0.2%Mo		1% Cr 0.20%Mo		1.85 %Ni, 0.8%Cr 0.25%Mo

- A water quenched AISI1040 may only hardened on surface whereas the other 4 low alloy steels (though less hardened) exhibit a much larger martensitic penetration toward the core.

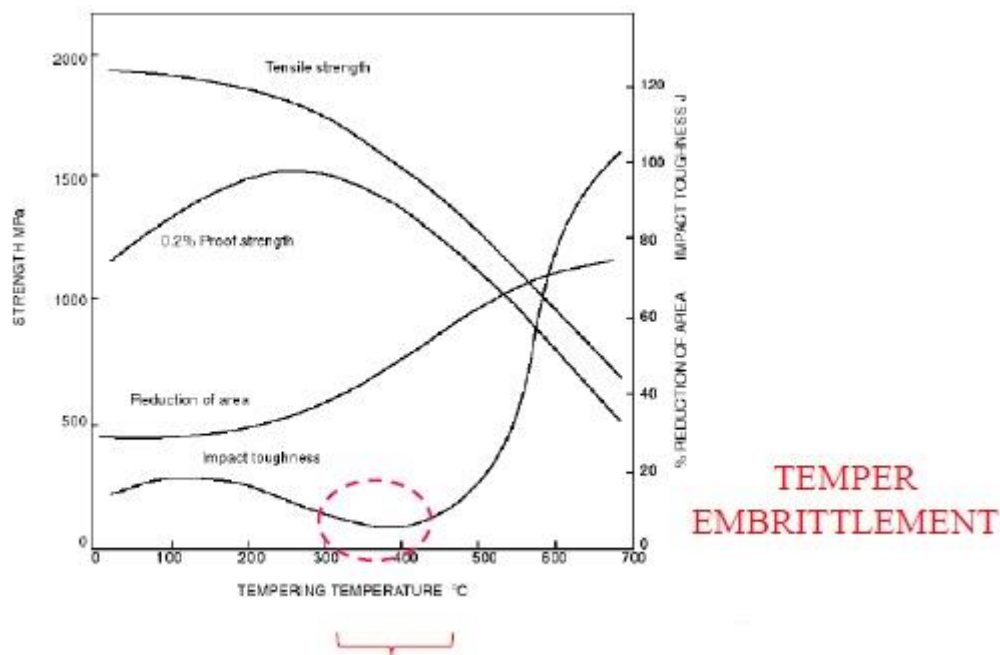
Hardness vs %elem (quenching medium) and Tempering temperature



We have different results from different steels.

Mechanical properties of steels after tempering: AISI 1040

Some steels may suffer embrittlement, which can be visualized by making a diagram with strength and toughness with varying temperature.



Decrease of toughness at a certain temperature range (350-450): lowest toughness → embrittlement.

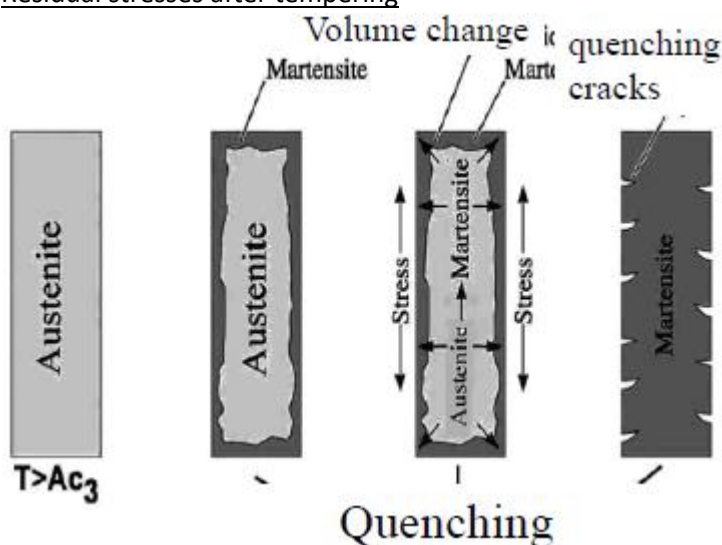
For low alloying steels, the embrittlement is not often observed, because we have always a small addition of Molybdenum in the steel. In plain carbon steels, it can be observed because there is no Molybdenum.

Effects of Alloying Elements on Steel

- **Manganese** contributes to strength and hardness; dependent upon the carbon content. Increasing the manganese content decreases ductility and weldability. Manganese has a significant effect on the **hardenability** of steel.
- **Phosphorus** increases strength and hardness and decreases ductility and notch impact toughness of steel. The adverse effects on ductility and toughness are greater in quenched and tempered higher-carbon steels.

- **Sulfur** decreases ductility and notch impact toughness especially in the transverse direction. Weldability decreases with increasing sulfur content. Sulfur is found primarily in the form of sulfide inclusions.
- **Silicon** is one of the principal deoxidizers used in steelmaking. Silicon is less effective than manganese in increasing as-rolled strength and hardness. In low carbon steels, silicon is generally detrimental to surface quality.
- **Copper** in significant amounts is detrimental to hot-working steels. Copper can be detrimental to surface quality. Copper is beneficial to atmospheric corrosion resistance when present in amounts exceeding 0.20%.
- **Nickel** is a ferrite strengthener. Nickel does not form carbides in steel. It remains in solution in ferrite, strengthening and toughening the ferrite phase. Nickel increases the **hardenability** and impact strength of steels.
- **Molybdenum** increases the **hardenability** of steel. It enhances the creep strength of low-alloy steels at elevated temperatures.

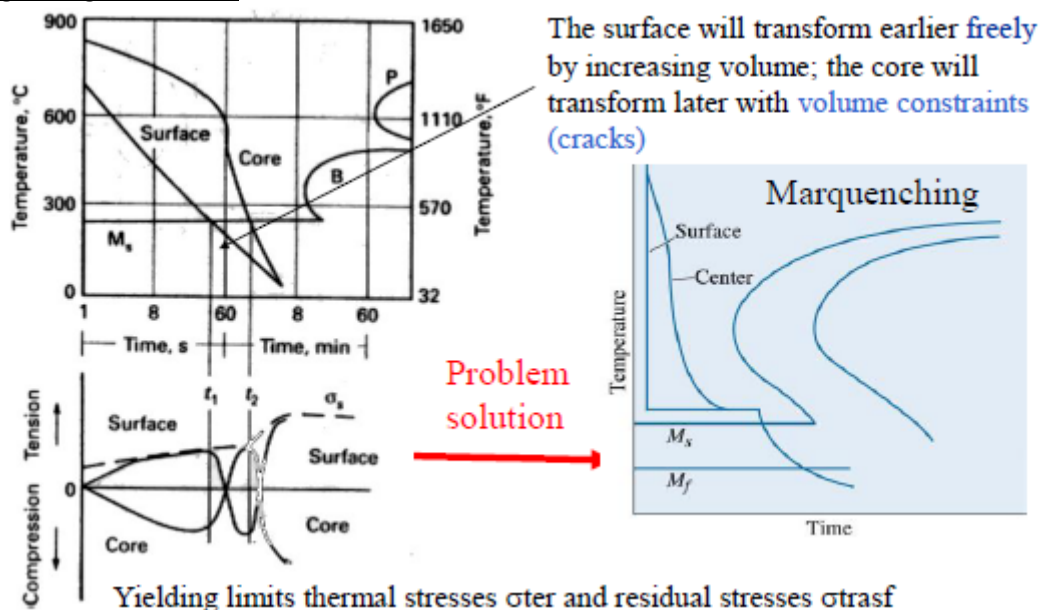
Residual stresses after tempering



The austenite-martensite transformation on quenching causes a volume change. As a result, quenching cracks originate due to phase transformation.

To avoid crack, we have to exploit the hardenability of steel and add appropriate alloying elements in order to cool more smoothly on the surface and allow the more gradual transformation of austenite, not only on the surface but also inside → homogeneous transformation of austenite from the surface to core → the material can expand without cracking.

Quenching of large diameters



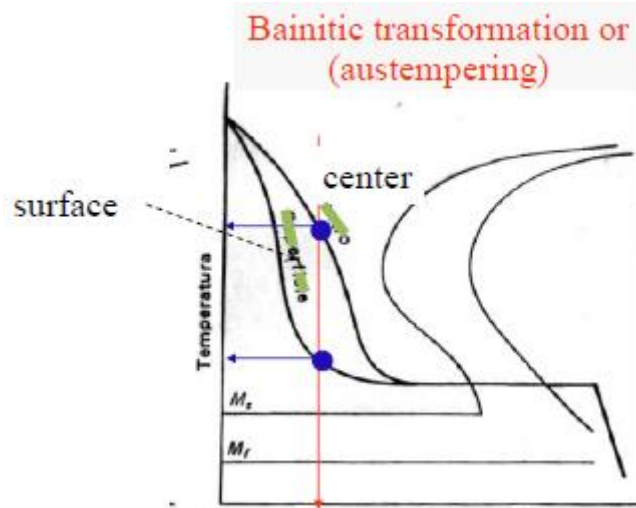
Marquenching: different behavior of surface and center; before obtaining martensite, we do a kind of thermal equilibrium → in the first part, we can cool separately surface and core but when we reach 450°C

just before martensite starts, we draw the material from quenching. The surface and the center will keep the same temperature.

Large diameters → surface and core follow different cooling curves (the core has a slower cooling curve) → cracks more likely to happen in the case of larger diameter. Tensile stress on surface and compressive in the core → crack can open and cross the overall part leading to failure.

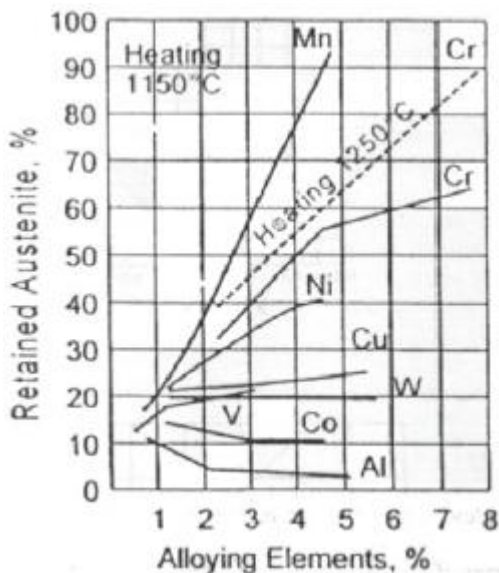
Quenching of large diameters: interrupted quenching or austempering

Purpose: interrupted cooling to avoid distortions and cracks.



Residual Austenite vs %element additions

If we want 100% martensite from quenching and instead we obtain residual austenite → Bad!



← Alloying elements which may promote the formation of residual austenite after quenching

Inconveniences on tempering: residual austenite

On quenching: the transformation $\gamma \rightarrow m$ may be incomplete, it occurs when $T_{\text{quench}} > M_f$. Note that $M_s = f(\% \text{elem dissolved into the matrix}) \rightarrow$ with $\uparrow \%C$ and other austenitizing alloy elem., $\uparrow \% \gamma_{\text{res}}$; with $\uparrow CR$, $\uparrow \% \gamma_{\text{res}}$; on **stepped quenching**, $\uparrow \% \gamma_{\text{res}}$, because the isothermal soaking stabilizes γ

On tempering: the γ content that can be transformed at room temperature depends on %alloy elem: the higher the content of alloy elements, the higher the tempering temperature, the larger fraction of residual γ .

Consequences of residual γ : $\downarrow$ HRC, Rfatigue, $\uparrow$ dimensional instability; internal stresses (γ transforms very slowly and determines volume changes).

Goal: the intent is not to completely eliminate γ , rather to impede the formation of metastable γ in the finished product; austenite imparts resilience, sometimes it can be useful; the only problem is that sooner or later it will transform into martensite or bainite.

Solution: execute **two or more tempering treatments** (e.g. carburizing steels); **quenching under-zero degree C** (e.g. -80°C with dry ice and acetone; up to -140 and -150°C , in liquid nitrogen) in order to surpass M_f + **relaxation** (e.g. medium C alloy steels).

Approximate distribution of alloying elements in alloy steels

Element	Dissolved in ferrite	Combined in carbide	Combined as carbide	Compound	Elemental
Nickel	Ni			Ni_3Al	
Silicon	Si				$\text{SiO}_2 \cdot \text{M}_x\text{O}_y$
Manganese	Mn $\longleftrightarrow$ Mn		$(\text{Fe}, \text{Mn})_3\text{C}$	$\text{MnS}; \text{MnO} \cdot \text{SiO}_2$	
Chromium	Cr $\longleftrightarrow$ Cr		$(\text{Fe}, \text{Cr})_3\text{C}$ Cr_7C_3 Cr_{23}C_6		
Molybdenum	Mo $\longleftrightarrow$ Mo		Mo_2C		
Tungsten	W $\longleftrightarrow$ W		W_2C		
Vanadium	V $\longleftrightarrow$ V		V_4C_3		
Titanium	Ti $\longleftrightarrow$ Ti		TiC		
Niobium	Nb $\longleftrightarrow$ Nb		NbC		
Aluminum	Al			$\text{Al}_2\text{O}_3; \text{AlN}$	
Copper	Cu (small amount)				
Lead	Pb				Pb

The arrows indicate the relative tendencies of the elements listed to dissolve in the ferrite or combine in carbides.

After E. C. Bain and H. W. Paxton, "Alloying Elements in Steel", ASM, 1966.

Applications of tempered steels

Use	Type of Steel
Body	
Body sheet metal	1006, 1008, SAE 950, HSLA*
Suspension and steering	
Struts	1040 and 1030
Pitman and idler arms	1038, 1040, and 1041
Torsion bar housing	1021
Steering knuckles	1046
Tie rod ends	1040
Ball joint studs	1041
Center link	1040
Engine	
Pulley	1010
Crankshaft	1046, 1049 modified, nodular cast iron
Connecting rod	1041
Piston pin	1016
Rocker arm	1040, 1010 case-hardened (stamped)
Intake valve	1041, 1547
Oil pan	1010

Applications of tempered steels

Torque converter: flex plate	1020 to 1050
Cover, turbine and impeller shell, and valves	1006 and 1010
Overrunning clutch cam	1060 modified
Overrunning clutch hub	1060
Impeller hub	1137
Transmission	
Input shaft	1024, 1045 induction hardened
Output shaft	1024, 1036, 1045 induction hardened, 4027 carburized
Kickdown and reverse bands	1040, 1050 quenched and tempered
Sun gear driving shell	1010
Planet pinion shaft	1041, 1050 induction-hardened, 8620 carburized
Clutch disks and plates	1020 to 1030

* High-strength low-alloy steels.

Lecture 13

Thermal and Chemical Surface Treatment of Steels

Hardening: the material will dissolve the carbon already contained in the steel and after austemizing, it will make diffusion uniform; after quenching with water on the surface then we make hardening. Material uses its own carbon to make the surface hard. The material must have at least 0,4% Carbon.

Hardness is very convenient for wear resistance and to withstand specific forces in a point in order to avoid denting of the surface.

Hardening by Carburizing

- **Purpose:** high hard surface and tough core

- **Exploited factors:**

- martensite increases its hardness with increasing %C
- martensite forms from austenite on quenching
- C is highly soluble in austenite
- absorption/ diffusion rate increases < with increasing T
- Process conditions: $T_c = 870+950^{\circ}\text{C}$ time = 10-20 h
- Applicable to carburizing steels (plain carbon steel <0.2% and low carbon low alloy steels)

Long time because
diffusion is a slow process

- **Applications:** gears, shaft, bearings. piston rod, etc. (**problems:** wear and fatigue)

- **Process outcome**

- Carbon concentration profile along the thickness (max HV (hardness) on surface)
- Transformation of austenite into martensite by oil quenching

Because diffusion gives a different rate with the depth, we may expect a decreasing hardness while increasing depth. Carbon will be enriched on the surface up to 0,8% → by waiting 20 hours we may expect carbon to diffuse into the thickness and have a certain profile.

Oil quenching (mild quenching) allows transforming austenite into martensite without cracking because 0,8%C is so high and martensite will be so hard that may crack if we use strong quenching.

Hardening by Nitriding

- **Purpose:** extremely hard surface (much harder than carburizing) and tough core

- **Exploited factors:**

- N with Fe form very hard compound (nitrides)
- Solubility of N in ferrite is very low
- At the stability temperature of ferrite, diffusion is slow
- the formation of nitrides inhibits further diffusion of N
- Process conditions: $T_N = 500-600^{\circ}\text{C}$ time = 70 h
- Applicable to nitriding steels (plain medium carbon steel <0.4- 0.6% and medium carbon low alloy steels in tempered/semifinished conditions)

Longer time →
diffusion into ferrite

- **Applications:** gears, valves, injection pump parts, high wear parts, high localized loaded parts

- **Process outcome:**

- Formation of very high hardness superficial nitrides ~ 1100 HV
- 0.1-0.6 mm penetration depth (thickness);

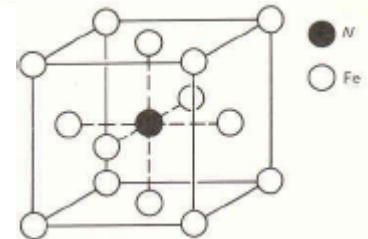
The advantage is that after nitriding we don't need any surface treatment.

In order to make nitriding, we need ammonia gas atmosphere which must be dissociated to obtain atomic nitrogen which will react with Fe, Al, and Cr to make a very hard nitride, Fe_4N .

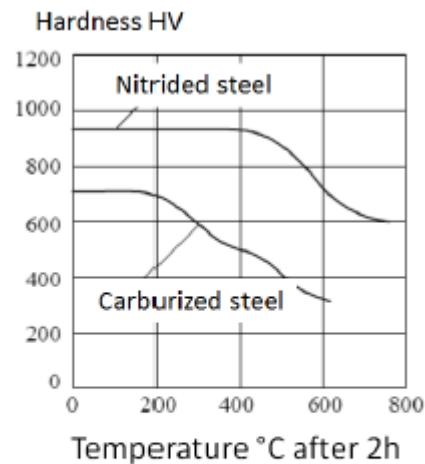
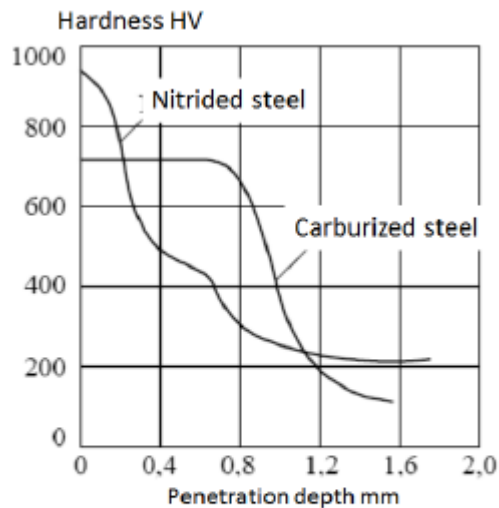
Nitriding steels

Sigla UNI	30 Cr Mo 10	42 Cr Al Mo 7	38 Cr Al Mo 7
%C	0,30	0,42	0,38
%Si	0,30	0,35	0,35
%Mn	0,60	0,55	0,60
%Cr	2,50	1,70	1,70
%Mo	0,40	0,25	0,35
%Al	—	0,40	1,0
Hardness after nitriding	650 HV	900 HV	1050 HV

Nitriding steels have some amount of aluminum already into them. Al combines with N to make very hard nitrides.



Carburizing vs Nitriding



Harris' formula

$$\text{Penetration depth (mm)} = 660 \cdot \sqrt{t} \cdot \exp\left(-\frac{8287}{T}\right) \quad \begin{matrix} T \text{ [K]} \\ t \text{ [h]} \end{matrix}$$

Nitrided steels: they have larger hardness, but it drops very sharply in few tens of mm (first graph) and after 400°C (second graph). Nitrided steels are much harder with a higher melting temperature, they are more stable at high T.

Carburized steels: the hardness is lower, but the thickness is larger (first graph). With lower time, we can increase the penetration depth of Carbon. Hardness drops dramatically after 200°C (second graph).